Subspace Circuit Discovery via Sparse Transformer Factorization

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Abstract

[1 sentence: Circuit discovery identifies which edges in a transformer's computational graph matter for a task, but says nothing about *what* information flows along each edge — which subspaces of the residual stream carry the signal.]

[1 sentence: We decompose each weight matrix as $W = SF$ where F is a shared factor bank and S is a sparse selector, trained by distillation to faithfully reproduce a pretrained model (GPT-2 Small, CE gap <0.02 nats at 99% sparsity).]

[1 sentence: This factorization makes existing circuit methods — EAP, EAP-IG, EAP-GP, DAS — automatically produce per-factor attribution scores: each edge score becomes a sum over per-factor contributions via a purely algebraic identity, yielding a 3D attribution tensor of shape (writers $\times$ readers $\times$ factors).]

[1 sentence: PCA on this 3D tensor (flattened to edges $\times$ factors) automatically decomposes circuits into interpretable sub-circuits — groups of edges sharing the same factor subspace — with auto-k at 90% variance requiring no per-task tuning; Tucker decomposition of the same tensor independently recovers identical factor directions (cosine similarity 1.000, Grassmannian geodesic distance ≈ 0), confirming these sub-circuits are real structure, not decomposition artifacts.]

[1 sentence: The factor bank also provides a geometric inductive bias for causal variable localization: constraining PCA to the factor subspace matches DAS at 98.8% IIA with zero training, and initializing DAS from this subspace improves hard-example IIA from 73.5% to 91.2%.]

[1 sentence (NEW v6): Analyzing the relationship between attribution and causal structure reveals a striking finding: the factor subspace explaining most attribution variance is nearly orthogonal ($\sim$88°) to the causally-active subspace (DAS), with a 14–40$\times$ gap between activation universality and causal specificity. Tucker modes capture the modulation component — necessary for task performance (removing 8 modes drops logit diff by 21%) but not sufficient in isolation (keeping only Tucker modes reverses the logit diff). The factor manifold acts as a gradient attractor (ρ: 26%$\rightarrow$83%), determining whether DAS training finds the global optimum — a launch pad for causal discovery, not a container.]

1 Introduction

[1 paragraph: Edge-level circuit discovery (EAP, ACDC) tells us *which* attention heads communicate for a task, but each edge is a single scalar. We don't know which dimensions of the 768-dimensional residual stream carry the signal, whether an edge transmits one signal or several superimposed ones, or whether two edges carry the same information. Causal variable localization (DAS) finds the relevant subspace but requires supervised training on counterfactual pairs — specifying the causal model in advance. There is no general-purpose method to decompose an edge into its constituent factor-level contributions.]

[FIGURE 1 — NEED: Three-panel pipeline figure. (A) Standard EAP circuit with scalar edges. (B) Factorized circuit with per-factor edge bundles. (C) PCA denoising: scree plot + CMD comparison bar chart.]

[1 paragraph: We propose *sparse weight factorization*: every weight matrix W in a pretrained transformer is decomposed as $W = SF$, where F is a shared overcomplete factor bank and S is a sparse per-projection selector, trained by distillation on the composite QK and OV circuits. The model computes through factors — $Wx = S(Fx)$ — so factors are part of the computation, not a post-hoc reconstruction. This structural property makes existing circuit methods automatically produce per-factor scores: factorized EAP decomposes each edge score into a sum over per-factor contributions via a purely algebraic identity. Factorized DAS identifies which factors span the causal subspace. The contribution is the factorized model and the adapted methods; PCA sub-circuit decomposition is the primary tool applied to the factor-level scores, with Tucker decomposition providing independent validation.]

[1 paragraph: Our main finding is that per-factor attribution reveals structure invisible to scalar edge scores. PCA on the per-factor attribution tensor automatically decomposes circuits into interpretable sub-circuits — groups of edges sharing the same factor subspace — and Tucker decomposition independently recovers identical factor directions (cosine similarity 1.000, Grassmannian geodesic ≈ 0, 8/8 top-edge match on IOI). This convergence of two mathematically different methods is strong evidence that the sub-circuits are real. PCA at auto-k (90% variance) requires no per-task tuning: IOI k=108, SVA k=5, GT k=79, Gender k=190, Capital k=29 — adapting to circuit complexity automatically. The right unit of analysis is the *factor subspace*: individual factor attribution is near chance (gauge-dependent), but the PCA/Tucker mode spans are gauge-invariant and causally meaningful. The factor bank also serves as a geometric inductive bias for causal variable localization: constrained PCA (restricting to the top 1% of factors) matches DAS at 98.8% IIA with zero training on regular examples, and initializing DAS from this subspace improves hard-example IIA from 73.5% to 91.2% despite starting 82.5° from the optimum. Factor-constrained subspaces]

[Contributions:

1. **Sparse weight factorization** ($W = SF$): a shared factor bank + sparse selectors, trained by composite-circuit distillation (QK, OV), achieving Pareto-optimal sparsity vs. fidelity across 7 selector mechanisms and bank sizes from 1024 to 40000. The factorized model is causally equivalent to the pretrained model (CE gap $<$0.02 nats, DAS IIA $\geq$0.97 bidirectional). (§3)

2. **Factorized circuit methods**: algebraically lossless per-factor decomposition of EAP edge scores, extended to EAP-IG, EAP-GP, and DAS. Each adapted method returns standard outputs plus per-factor attribution at zero additional cost. (§4; proof in Appendix B)

3. **Sub-circuit decomposition via factor subspaces**: PCA on the per-factor attribution tensor automatically decomposes any circuit into interpretable sub-circuits, each mediated by a distinct factor subspace (IOI: early heads $\rightarrow$ MLP0, S-inhibition fan-out, name movers $\rightarrow$ logits). Tucker decomposition independently recovers identical factor directions (cosine 1.000, Grassmannian geodesic ≈ 0), confirming the sub-circuits are real structure, not decomposition artifacts. The factor *subspace* is gauge-invariant and causally meaningful; individual factors are not. (§5)

4. **Causal variable localization via the factor manifold**: Constraining PCA to factor subspace matches DAS on regular examples (98.8% IIA, zero training). Manifold-initialized DAS achieves 91.2% hard IIA vs 73.5% vanilla. (§4)

5. **Attribution-causal orthogonality and the background/modulation decomposition** (NEW v6): The factor subspace explaining most attribution variance is $\sim 88°$ from the causally-active subspace across 5 tasks, with a 14–40× activation-causal gap. Tucker C modes capture the modulation component — necessary (removing 8 modes drops logit diff 21%) but not sufficient (keeping only modes reverses logit diff to −30%). Different corruptions route through partially distinct factor subspaces (principal angles 58–78°). Each circuit pathway uses nearly disjoint factor subsets (cross-pathway Jaccard < 0.5%). (§7)

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2 Background

2.1 Edge Attribution

[3-4 sentences: Attribution patching approximates activation patching via first-order Taylor expansion, giving per-module importance in O(1). EAP (Syed et al. 2024) extends this to edges — the connection from writer u to reader v through the residual stream. State the EAP formula: $\Delta L_{uv} = \sum_n \langle r_u'^{(n)} - r_u^{(n)},\ \partial L / \partial \text{resid}_v^{(n)} \rangle$. Each edge gets one scalar: important, but no information about *what* flows.]

2.2 Causal Abstraction

[3-4 sentences: Geiger et al. (2024) frame interpretability as alignment between a neural network and a causal model. DAS learns a rotation R that aligns the causal variable with the first k axes, then swaps those k dimensions between inputs. Interchange Intervention Accuracy (IIA): fraction where the patched model predicts the source answer. This identifies *where* information is encoded but requires specifying the causal model and training R.]

2.3 MIB Evaluation Framework

[3-4 sentences: MIB (Mueller et al. 2025) standardizes evaluation. Track 1 measures circuit faithfulness at multiple sizes via activation patching; CPR $= \int_0^1 f(\mathcal{C}_k)\, dk$ (area under faithfulness curve, higher = better) and CMD $= \int_0^1 |1 - f(\mathcal{C}_k)|\, dk$ (distance from perfect reproduction, lower = better). State the faithfulness formula: $f(\mathcal{C}, \mathcal{N}; m) = (m(\mathcal{C}) - m(\varnothing))/(m(\mathcal{N}) - m(\varnothing))$.]

3 Sparse Weight Factorization

3.1 Architecture

[Bolded opener: **Every weight matrix in the model can be faithfully decomposed as $W_i = S_i F$, where F is a shared overcomplete factor bank and S_i is a sparse selector, achieving 99% sparsity with no CE degradation.**]

[1 paragraph: $F \in \mathbb{R}^{N_F \times d_m}$ is a factor bank whose rows are directions in the residual stream, shared across all projections $(W_Q, W_K, W_V, W_O, W_{\text{in}}, W_{\text{out}})$. $S_i \in \mathbb{R}^{d_h \times N_F}$ is a per-projection sparse selector. The bank is overcomplete ($N_F \geq d_m$; we use $N_F = 8192$ for GPT-2's $d_m = 768$). Each channel in each head activates only a small subset of factors, so the selector matrix is highly sparse — analogous to an SAE's encoder activations, but in *weight space*. The key difference from SAEs: factors are part of the model's computation (the model literally computes $W_i x = S_i F x$), not a post-hoc decomposition of what the model outputs.]

3.2 Distillation on Composite Circuits

[1 paragraph: We distill on the *composite* circuits $QK_i = W_Q^i (W_K^i)^\top$ and $OV_i = W_V^i W_O^i$, not on individual weight matrices. This is necessary because of rotational gauge freedom: for any orthogonal R, replacing $W_Q \to W_Q R$ and $W_K \to W_K R$ leaves the QK circuit unchanged but produces a completely different individual factorization. The composite circuits are gauge-invariant observables; matching them is both necessary and sufficient for faithful reproduction. Formal statement and proof in Appendix A.]

[2-3 sentences: Core loss: MSE on QK and OV circuits summed over layers. Sparsity via selector mechanism (DST or JumpReLU with straight-through estimator). Only F and S optimized; all other parameters copied from the pretrained model. Full training details in Appendix E.]

3.3 Model Equivalence

[Bolded opener: **The factorized model is not merely an approximation of GPT-2 — it is causally equivalent. CE gap <0.02 nats, DAS IIA ≥0.97 bidirectional on IOI, and MAS confirms the factorized model's representations are interchange-compatible with GPT-2's.**]

[1 paragraph: Three levels of equivalence evidence: (1) **Behavioral**: CE gap <0.02 nats at 99% selector sparsity. The factorized model produces the same next-token distribution as GPT-2. (2) **Representational**: DAS trained on GPT-2 representations achieves IIA ≥0.97 when applied to the factorized model's representations, and vice versa. This bidirectional interchange (MAS) confirms the internal representations are structurally aligned, not just input-output equivalent. (3) **Circuit-level**: EAP-IG on the factorized model recovers the same circuit structure as on GPT-2 (same top edges, same head rankings). The factorized model eliminates negative name movers — heads that exist in GPT-2 to cancel other heads' errors — because the factorized representation doesn't need that cancellation structure.]

3.4 Sparsity–Fidelity Tradeoffs

[FIGURE 2 — NEED: Pareto frontier. CE vs L0/channel by selector type. DST reaches GPT-2 CE at ∼100 active factors/channel (99% sparse). Horizontal line at GPT-2 baseline CE.]

[2 sentences: DST achieves Pareto-optimal tradeoffs: 99% sparsity with <0.02 CE gap from GPT-2 (∼3.25). The overcomplete factor bank ($N_F = 8192 \gg d_m = 768$) enables extreme sparsity without information loss.]

4 Factorized Circuit Methods

[Bolded opener: **The $W = SF$ structure makes four existing circuit methods — EAP, EAP-IG, EAP-GP, and DAS — automatically produce per-factor attribution scores via algebraic decomposition, requiring no additional forward passes.**]

4.1 Factorized EAP and EAP-IG

[1 paragraph: We first state standard EAP (Eq. **??**): the edge score ΔL_{uv} between writer u and reader v is the sum over tokens of the writer's output difference dotted with the reader's gradient. Show the formula with shape annotations (as in Ivan's draft §2.1). Then substitute $W_u^{o\top} = S_u F$ and $W_v^{in} = F^\top S_v^\top$, expand, and rearrange to get $\Delta L_{uv} = \sum_{i,j} \Delta L(u, i, v, j)$ (Eq. **??**).

The per-factor score $\Delta L(u, i, v, j)$ measures how much writing by module u into factor i, as read by module v through factor j, affects the loss. This decomposition is exact — purely algebraic, no approximation. We present EAP (not EAP-IG) because the linear approximation makes the algebra exact; EAP-IG uses integrated gradients and the decomposition becomes approximate. Full derivation with LayerNorm Jacobian expansion in Appendix B.]

[1 paragraph: State the losslessness formally as a proposition. **Proposition 1** (Factorized EAP aggregates to EAP): $\Delta L_{uv} = \sum_{i=0}^{N_F} \sum_{j=0}^{N_F} \Delta L(u, i, v, j)$. This is immediate from the derivation — factorized EAP is strictly more informative than standard EAP at zero additional cost (one forward + one backward pass, same as standard EAP). The closed-form score $\Delta L(u, i, v, j) = M_{ij} T_1^{(uv)} - \frac{1}{d_m} T_2^{(uv)}$ (where M is the factor Gram matrix and T_1, T_2 are sum-of-outer-products) enables efficient computation via einsums. Closed-form derivation in Appendix C.]

4.2 Factorized EAP-GP (GradPath)

[1 paragraph: EAP-GP (gradient path attribution) tracks gradients through the full computational graph. The $W = SF$ substitution applies identically: each GradPath edge score decomposes into per-factor contributions. Brief comparison of EAP-GP CMD vs EAP-IG CMD on IOI.]

4.3 Factorized DAS and Causal Subspace Localization

Standard IIA on regular examples is a deceptively easy metric: zero-training constrained PCA achieves 98.8%, trained DAS achieves 94–97%, and even random baselines exceed 90% because fewer than 2% of examples are flippable. Hard examples (logit margin < 1.0) are the only regime that differentiates methods. On hard examples, CPCA-initialized DAS achieves the best results of any initialization strategy (91.2% at $k = 1$, 97.1% at $k = 4$, and 100% strict IIA at $k = 4$), while random-init DAS reaches only 73.5%. Thirteen untrained methods and five structured-basis geodesic searches all fail to exceed the majority-class baseline (70.6%) on hard examples, confirming that training is required — but the factor manifold determines *where* on the Grassmannian that training succeeds.

4.3.1 Regular Examples: CPCA Matches Trained DAS

Standard DAS learns a rotation R in $\mathbb{R}^{d_m}$. Factorized DAS constrains R to the span of the factor bank via $U = F^\top A$ where $A \in \mathbb{R}^{N_F \times k}$ is sparse (group lasso penalty). But a simpler approach works for regular examples: *constrained PCA* (CPCA) selects the top ∼1% of factors by selector magnitude (∼81 of 8192), projects activation deltas into this 81-dimensional subspace, and takes the top-k principal components. This zero-training method achieves 98.8% IIA on IOI at Layer 9 — compared to 21.3% for unconstrained PCA at Layer 8. The factor bank collapses the search space from $\mathrm{Gr}(k, 768)$ to $\mathrm{Gr}(k, 81)$, making PCA sufficient where it normally fails.

This near-perfect regular IIA is *not* evidence that the causal variable is easy to find — it reflects the metric's insensitivity. On the regular evaluation set, only 2/370 IOI and 5/380 SVA examples are flippable, so any reasonable direction achieves ≥99% IIA. Hard examples are where methods actually separate.

4.3.2 Hard Examples: CPCA-Init DAS is the Best Strategy

On hard examples (logit margin < 1.0, ∼8% of data), CPCA alone achieves only 47–53% IIA. But *initializing* vanilla DAS from the CPCA direction — then training in unconstrained $\mathbb{R}^{768}$ — yields the best performance of any initialization strategy: 91.2% hard IIA at $k = 1$ (100 steps), versus 73.5% for random-init DAS. At $k = 4$, CPCA-init DAS achieves 97.1% IIA and **100% strict IIA** on both IOI and SVA, while unconstrained PCA init achieves only 66.7%/90% strict IIA. The CPCA direction is 82.5° from the DAS optimal (nearly orthogonal on $\mathrm{Gr}(1, 768)$) — the factor manifold provides a better *optimization basin*, not a better initial direction. More constraint is worse: Riemannian DAS (Stiefel manifold constraint) drops to 64% on regular IOI. The factor manifold is a launch pad, not a prison.

Table 1: DAS IIA on hard IOI examples (34 eval, Layer 8). CPCA-init unconstrained achieves the best result across all four conditions.

	Unconstrained	**On-manifold (Riemannian)**
Random init	73.5%	88.2%
CPCA init	**91.2%**	64.0%

4.3.3 Riemannian Gradient Decomposition

Why does CPCA initialization outperform all alternatives? We decompose each DAS gradient step into components tangent and normal to the factor manifold $\mathrm{Gr}(1, 81) \subset \mathrm{Gr}(1, 768)$. The manifold gradient fraction $\rho = \|\nabla_{\text{on}}\|/\|\nabla_{\text{tang}}\|$ measures how much useful gradient signal lies along the factor manifold.

Two findings explain CPCA-init's superiority: (1) ρ *rises* from 26% to 82.5% for CPCA init, while staying flat at ∼50% for unconstrained PCA init — the factor manifold is a *gradient attractor* (Figure 1). (2) CPCA init converges to the global optimum (0.1° away); unconstrained PCA init converges to a local optimum 23.8° away, despite starting 17° closer. On $\mathrm{Gr}(4, 768)$, the effect is stronger: CPCA init converges to 0.59° from optimal while unconstrained PCA gets stuck at 86.5°.

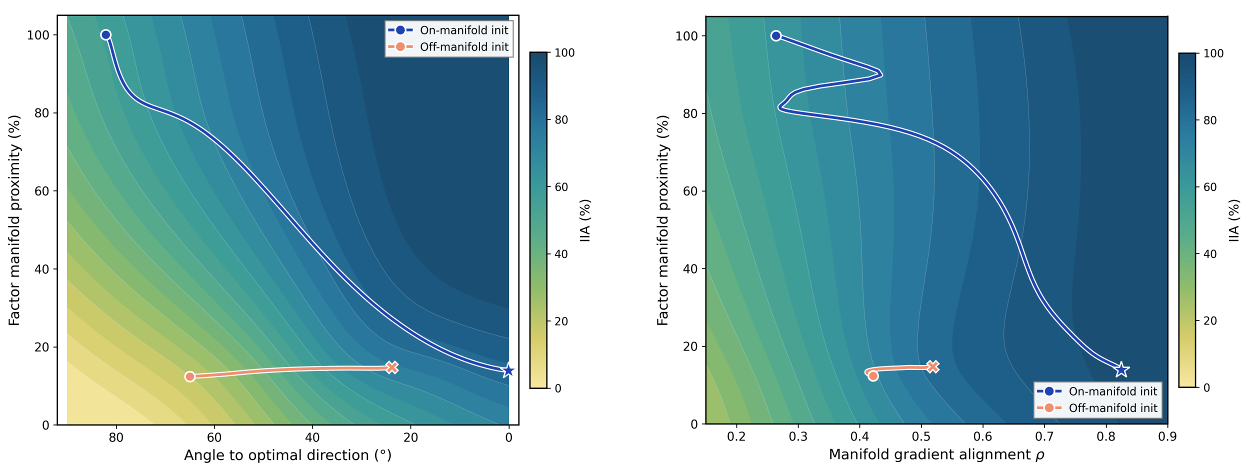

Figure 1: **Grassmannian landscape for DAS training on hard IOI,** $\mathrm{Gr}(1, 768)$**.** *Left:* IIA as a function of angle to the DAS optimal direction (x-axis) and factor manifold proximity (y-axis). The on-manifold init (blue, starting at 100% proximity) follows a high-IIA ridge to the global optimum (star); the off-manifold init (orange, starting at ∼12% proximity) gets trapped in a local optimum. *Right:* Same landscape, x-axis replaced by manifold gradient fraction ρ. The on-manifold trajectory's ρ rises from 0.26 to 0.83 (gradient attractor); the off-manifold trajectory stays flat at ∼0.35.

4.3.4 Geodesic Line Search: Training-Free Methods Fail on Hard Examples

Can we find high-IIA directions without gradient descent? We implement a geodesic line search: Riemannian gradient at Q_0, 30-point coarse sweep along the geodesic, 10-step ternary refinement ($\sim$40 evaluations total, vs. 200 gradient steps for DAS). We compare five initialization bases — factor bank (CPCA), SAE decoders, unconstrained PCA, probing directions, and random — each with 10–11 starts.

Table 2: Geodesic line search IIA on hard IOI (34 eval, Layer 8). Mean ± std over 10–11 runs. The 70.6% floor (= 24/34) is the majority-class baseline.

Basis	**Mean (k=1)**	**Max**	**Mean (k=4)**	**Max**
Probing	81.3 ± 2.0	82.4	80.5 ± 3.0	82.4
Unconstrained PCA	72.7 ± 7.1	94.1	73.3 ± 7.9	97.1
Factor bank (CPCA)	70.6 ± 0.0	70.6	70.9 ± 0.9	73.5
SAE decoders	70.6 ± 0.0	70.6	71.1 ± 1.2	73.5
Random	70.9 ± 0.9	73.5	70.6 ± 0.0	70.6
Trained DAS (CPCA init)	*94.1*		*97.1*	

A single geodesic step is insufficient: nearly all bases converge to the majority-class baseline (70.6% = 24/34). Probing, which is supervised, reaches 81% but cannot flip the hard examples. The factor bank (CPCA) is trapped at the baseline from every start — the geodesic cannot escape the local landscape. This is why CPCA *initialization* for trained DAS works while CPCA alone does not: the factor manifold identifies the right basin, but reaching the optimum within that basin requires gradient descent through the IIA objective.

4.3.5 Comprehensive Untrained-Method Comparison

We further test 8 zero-training or low-training methods: CPCA baseline, difference-in-means, top EAP factors (gradient-weighted attribution), logit gradient (1 backward pass), constrained probing (200 steps in factor subspace), LDA in factor subspace, hard-example CPCA (PCA on hard deltas only), margin-weighted CPCA, and iterative geodesic descent (5 Riemannian steps from CPCA).

Table 3: Hard-example IIA for untrained methods vs. trained DAS. All factor-constrained methods have manifold fraction ≈ 1.0. No untrained method exceeds the majority-class baseline (70.6%).

Method	**Hard IIA (k=1)**	**Hard IIA (k=4)**
Iterative geodesic	70.6	70.6
LDA (factor subspace)	70.6	67.6
Constrained probing	67.6	64.7
Top EAP factors	64.7	47.1
Logit gradient	50.0	52.9
Margin-weighted CPCA	52.9	50.0
CPCA baseline	52.9	47.1
Hard-example CPCA	52.9	47.1
Difference-in-means	52.9	47.1
Trained DAS (CPCA init)	**91.2**	**97.1**
Trained DAS (random)	73.5	—

No untrained method exceeds the majority-class baseline (70.6%). The 70.6% barrier corresponds to 24/34 examples that are already correctly predicted; the remaining 10 require the intervention to *flip* the model's prediction. At 70.6%, none of these 10 are flipped. Only trained DAS breaks through, and CPCA-init DAS breaks through the farthest.

Hard-example CPCA (training PCA only on hard deltas) provides no benefit over standard CPCA — the hard examples do not occupy a spectrally distinct subspace. They differ in *degree* (smaller logit margin), not *kind* (different direction).

Takeaway. Regular IIA is not the right metric for evaluating causal variable discovery — it is too easy, saturating at $\geq$94% for any reasonable method. Hard examples are where methods differentiate, and on hard examples CPCA-initialized DAS is the best strategy we tested: 91.2% at $k = 1$, 97.1% at $k = 4$, and 100% strict IIA at $k = 4$. The factor manifold provides a 20× search space reduction ($\mathrm{Gr}(k, 81)$ vs. $\mathrm{Gr}(k, 768)$), places initialization in the global optimum's basin of attraction (Figure 1), and acts as a gradient attractor during training (ρ rises from 26% to 83%). It cannot replace training — but it determines whether training finds the right answer.

4.4 The Per-Factor Attribution Tensor

[1 paragraph: For circuit ranking, we marginalize over the reader factor to get a matrix $\mathrm{scores}_W \in \mathbb{R}^{n_{\mathrm{edges}} \times N_F}$ (69,865 × 8,192 for GPT-2 IOI). Each row is an edge's factor-attribution profile. Standard EAP is the row-sum (collapsing factor dimension). Factorized EAP preserves the full factor profile per edge. Or keep the 3D tensor $\mathbb{R}^{n_w \times n_r \times N_F}$ for Tucker decomposition. The question is: what is the intrinsic dimensionality of this factor-attribution space?]

5 Factor-Level Circuit Analysis: IOI Case Study

[1 paragraph: We demonstrate what per-factor attribution reveals using IOI as the primary worked example, with PCA decomposition across 5 tasks (IOI, SVA, Greater-Than, capital-country, gender bias). PCA on the flattened attribution tensor automatically decomposes circuits into sub-circuits: groups of edges sharing the same factor subspace. Each principal component defines a "factor direction" — edges loading on the same PC use the same factor subspace to communicate, forming a coherent sub-circuit. Tucker decomposition of the full 3D tensor independently recovers identical factor directions (cosine similarity 1.000, 8/8 top-edge match), confirming this structure is real. All use checkpoint atomic-sweep-40 (8192 factors, DST selector).]

5.1 PCA Sub-Circuit Decomposition

[Bolded opener: **PCA on the flattened attribution tensor $\mathrm{scores}_W \in \mathbb{R}^{(n_w \cdot n_r) \times N_F}$ automatically decomposes any circuit into interpretable sub-circuits, each mediated by a distinct factor subspace, with auto-k at 90% explained variance requiring no per-task tuning.**]

[1 paragraph: The factorized attribution tensor $\mathrm{scores}_W \in \mathbb{R}^{n_w \times n_r \times N_F}$ assigns each edge a vector in $\mathbb{R}^{N_F}$ rather than a scalar. Flattening to $\mathbb{R}^{(n_w \cdot n_r) \times N_F}$ and applying truncated SVD yields principal components — orthogonal directions in factor space — and per-edge loadings on each component. Edges loading on the same PC share a factor subspace: they communicate through the same subset of factors and form a coherent sub-circuit. The number of components k is set automatically at the 90% variance threshold: IOI k=108, SVA k=5, GT k=79, Gender k=190, Capital k=29. This requires no per-task tuning and adapts to circuit complexity — structurally simple circuits (SVA) concentrate variance in few components while complex circuits (Gender, IOI) spread it across many.]

5.2 Tucker Decomposition Recovers Identical Factor Directions

[Bolded opener: **Tucker decomposition of the full 3D tensor scores**$_W \approx \mathcal{G} \times_1 U_w \times_2 U_r \times_3 U_f$ **recovers factor modes U_f that are identical to the PCA principal components (cosine similarity 1.000, Grassmannian geodesic distance $\approx$ 0, 8/8 top-edge match), providing independent validation that the sub-circuit structure is real.**]

[1 paragraph: Tucker decomposition preserves the 3D structure of the attribution tensor, yielding writer modes U_w, reader modes U_r, factor modes U_f, and a core tensor $\mathcal{G}$. The factor modes $U_f \in \mathbb{R}^{N_F \times r_3}$ identify directions in factor space — the same object PCA finds from the flattened matrix. On IOI with Tucker ranks $(10, 10, 8)$: each of the 8 Tucker factor modes aligns with a PCA PC at $|\cos\theta| \geq 0.998$; the cross-method cosine similarity matrix is a near-perfect identity matrix (all off-diagonal entries < 0.04); and every matched pair identifies the same dominant edge. This convergence of two mathematically different methods — PCA maximizes variance of the flattened matrix, Tucker minimizes reconstruction error of the 3D tensor — is strong evidence that the factor sub-circuits reflect genuine structure in the attribution tensor, not artifacts of the decomposition. Tucker's additional writer and reader modes are fully determined by the (shared) factor modes: each factor direction is associated with a single dominant writer-reader pattern, so the extra dimensions add no information beyond what the PCA edge loadings already contain.]

[FIGURE — PCA vs Tucker factor direction agreement. 8×8 $|\cos|$ matrix = perfect diagonal. Geodesic distance matrix = all zeros on diagonal, $\pi/2$ off-diagonal. One compact panel suffices; no side-by-side comparison needed since the methods are identical. See pca_tucker_geodesic_agreement.png.]

5.2.1 Method and Rank Selection

[1 paragraph: We compute Tucker decomposition via Higher-Order SVD (HOSVD) initialization followed by Alternating Least Squares (ALS) refinement, using tensorly. Ranks (r_1, r_2, r_3) are selected by two criteria: (1) **variance elbow**: explained variance saturates at $(10, 10, 8)$ — increasing to $(15, 15, 8)$ adds <2% variance, while $(5, 5, 4)$ loses 20% (Figure **??**); (2) **core tensor concentration**: at $(10, 10, 8)$, only 5 of 800 core entries account for 50% of total energy, and 25 entries for 90%. Higher ranks add near-zero entries without improving the decomposition. The core tensor's extreme sparsity is the rank justification: the circuit is dominated by a handful of writer-reader-factor interactions, and the decomposition has captured all of them.]

[FIGURE 3 — NEED: Two-panel rank justification. Left: Variance explained vs Tucker rank config — (5,5,4), (8,8,6), (10,10,8), (12,12,8), (15,15,8), (20,20,16). Elbow at (10,10,8). Right: Cumulative core tensor energy vs number of entries (sorted by magnitude). At (10,10,8): 5 entries = 50%, 25 = 90%, 800 total. Shows extreme sparsity.]

5.2.2 A Priori Rank Selection via Effective Rank

[Bolded opener: **The effective rank per mode — computed by SVD on each unfolding of the attribution tensor before Tucker — predicts both the optimal Tucker configuration and whether Tucker will improve circuit scoring.**]

[1 paragraph: Before running Tucker, we compute the effective rank of each mode by SVD on the mode-n unfolding, measuring the number of singular values needed for 90% of the energy. This gives an a priori diagnostic:]

[TABLE 1a — Effective rank per mode across tasks.]

[1 paragraph: SVA's reader effective rank is 1 — a single reader mode (m0, MLP layer 0) captures 90% of the reader-mode energy, with an σ_1/σ_2 ratio of 8.0. This means the reader dimension is trivial: all circuit edges route through a single downstream target, and Tucker's reader decomposition reduces to a scalar multiplier that cannot change edge rankings. IOI (reader rank 17) and GT (reader rank 10) have multi-dimensional reader structure that Tucker can exploit. The effective rank table is both a rank-selection guide (set r_n at or above the 90% effective rank) and a diagnostic of circuit complexity: rank-1 modes indicate structurally simple circuits where the corresponding dimension offers no denoising opportunity.]

5.2.3 Scoring Results Across Tasks

[Bolded opener: **PCA at auto-k (90% variance) improves or matches EAP-IG CMD on 4/5 tasks with zero tuning. Tucker matches PCA's factor directions exactly but requires per-task rank selection to achieve its best CMD, and adaptive rank selection (variance-threshold) fails catastrophically.**]

[TABLE 1b — PCA and Tucker scoring vs standard EAP-IG.]

[1 paragraph: PCA with auto-k at 90% variance is the universal scoring method: it improves CMD on IOI (+34%), GT (+40%), Gender (+27%), and SVA (+5%) with no per-task tuning. Tucker at its best fixed ranks matches or exceeds PCA on IOI (+59%) and GT (+39%), but requires per-task rank selection — there is no universal Tucker configuration. Critically, adaptive Tucker (setting ranks to mode effective ranks at 50–90% variance) is catastrophically worse than EAP-IG (CMD 10–27× higher), because Tucker's scoring value comes from aggressive compression/denoising, not faithful reconstruction. PCA operates at 90% variance (gentle denoising); Tucker must operate at ∼50–60% variance (aggressive denoising) to improve scoring, but the optimal aggressiveness varies by task. PCA's simplicity and universality make it the recommended default; Tucker's value is as an interpretability tool (§5.2), not a scoring method.]

[TABLE 1c — NEED: Combined table with effective rank, optimal Tucker config, CMD, core tensor sparsity across all 5 tasks.]

5.2.4 Interpretable Sub-Circuit Modes

[Bolded opener: **PCA automatically decomposes IOI into three functional groups — corresponding to known sub-circuit components — without any supervision, circuit labels, or causal model specification.**]

[1 paragraph: The 8 principal components of the IOI attribution tensor separate into three clean functional groups:

- **PC0–PC3 (47% variance):** Four L0 attention heads (a0.h1, a0.h4, a0.h5, a0.h3) writing to MLP layer 0 (m0). Each head occupies its own PC — they use *different factor subspaces* to write to the same reader, which is why they separate into 4 components. These are the duplicate-token and early-processing heads from Wang et al. (2023).

- **PC4–PC6 (11% variance):** a5.h5 (the S-inhibition head) fanning out to mid-layer readers: m5, a6.h3.Q, m6, a7.h0.K, a7.h11.Q, a8.h10.Q, a8.h11.K. Different PCs capture different reader targets of the same writer, with different factor subspaces mediating each pathway.

- **PC7 (2% variance):** Name mover heads (a9.h9, a9.h6, a10.h7, a11.h10) writing to logits. The final output circuit.

This is the core finding: **IOI is not one circuit — it is a composition of several sub-circuits, each mediated by a distinct factor subspace.** The factor bank provides the decomposition; PCA merely reveals it.

Also: zoomed heatmap showing top 15 edges per PC with head names and numeric loadings. See pca_subcircuit_zoomed.png.

Also: per-task versions with 8 PCs each, showing that different tasks have different layer structure. See pca_layer_heatmap_{task}.png.]**[FIGURE 4 — HAVE: Layer-aggregated PCA sub-circuit heatmap. 8 panels, one per PC. Each panel is a 13×13 (writer layer × reader layer) heatmap showing where that PC's energy is concentrated. PC0–3 light up embed/L0 → L0. PC4–6 light up L5 → L5/L6/L7. PC7 lights up L9/L10/L11 → logits. Clean separation of the three functional groups. See pca_layer_heatmap.png.**

Also: zoomed heatmap showing top 15 edges per PC with head names and numeric loadings. See pca_subcircuit_zoomed.png.

Also: per-task versions with 8 PCs each, showing that different tasks have different layer structure. See pca_layer_heatmap_{task}.png.]

[1 paragraph: The layer-aggregated view (Figure 4) collapses the 157×445 edge tensor to 13×13 (writer layer × reader layer) per PC, making the sub-circuit structure immediately visible. Each PC lights up a different layer pair: PC0–3 concentrate in embed/L0 → L0 (early processing), PC4–6 in L5 → L5–L8 (S-inhibition fan-out), PC7 in L9–L11 → logits (output). This decomposition requires no supervised labels, no causal model specification, and no additional forward passes — only SVD on the attribution matrix.]

5.2.5 Each Edge Uses a Distinct Factor Subset

[Bolded opener: Circuit stages use nearly disjoint factor subsets. Even edges within the same pathway share only 1.8–3.7% of their top-20 factors, and cross-pathway sharing drops to <0.5%.]

[1 paragraph: For each of the four canonical IOI pathways (Dup→Ind, Prev→Ind, Ind→SInh, SInh→NM), we pool the top-20 factors per edge (by attribution magnitude) and compute pairwise Jaccard similarity. Within-pathway Jaccard ranges from 0.018 (Dup→Ind) to 0.037 (Ind→SInh). Cross-pathway Jaccard is 0.001–0.004. Each edge operates through its own ∼20 factors out of the 8192-dimensional bank. This is not cherry-picking: the pattern holds across all 6 pathway pairs. Even edges doing the *same job* (e.g., two S-inhibition→name-mover connections) use almost entirely different factors — each edge has a unique factor fingerprint.]

[FIGURE — HAVE: 4×4 factor overlap heatmap across IOI circuit stages. Diagonal (within-pathway) = 0.018–0.037. Off-diagonal (cross-pathway) = 0.001–0.004. YlOrRd colormap, values 0–0.04. See fig3_factor_overlap_stages_v2.png in BATCH2/. Caption should explain: diagonal values are NOT self-similarity (which would be 1.0) but mean pairwise Jaccard between different edges within the same stage.]

5.2.6 Per-Edge Factor Fingerprints and Q/K/V Factor Sharing

[1 paragraph: Per-edge factor fingerprints confirm the disjointness finding at individual edge resolution: the top-20 factors for each Ind→SInh edge are largely non-overlapping, with different edges using different factor channels to carry information

between the same head pairs. Q/K/V projections within a single head use partially overlapping but not identical factor sets — Q and K share more factors with each other than with V, consistent with the $QK^\top$ attention-score duality observed in the 4D Tucker decomposition (§J.6).]

[FIGURE — HAVE: Left: Per-edge factor fingerprints for the 8 Ind→SInh edges (top-20 factors each, binary heatmap showing which factors each edge uses). Right: Q/K/V factor sharing for 4 IOI heads with overlap stats. See fig1_per_edge_factor_fingerprints.png and fig2_qkv_factor_sharing.png in BATCH2/.]

5.2.7 Cross-Task Factor Disjointness

[1 paragraph: At k=10, 76% of the most-attributed factors are task-disjoint (38 of 50 unique across 5 tasks). Per-edge Jaccard is 0.007 at k=10, with 89% of edges having zero factor overlap between task pairs. Tasks use the same factor bank but activate different subsets. This is consistent with the cross-pathway disjointness within IOI: the factor bank provides a large shared vocabulary, but each circuit edge — within or across tasks — selects its own sparse subset.]

5.2.8 Tucker on Pre-Identified Circuits

[1 paragraph: Tucker can also be applied to the *circuit sub-tensor* — the top-k% edges pre-identified by standard EAP-IG. On the top-1% sub-tensor (698 edges), Tucker (10,10,8) explains 93% variance on IOI, 95% on SVA, 79% on GT. The core tensor is sparse on all three tasks: 3–5 entries for 50% energy. Writer and reader modes are near-axis-aligned (loading >0.99 on individual heads) across all tasks. *This confirms that Tucker's modular decomposition is a general property of circuit sub-tensors, not an IOI artifact.* However, the interpretable sub-circuit structure (a5.h5 → a9.h9) only emerges for IOI; SVA and GT are dominated by infrastructure pathways (layer-0 heads → MLP0).]

5.2.9 Q/K/V Separation Is Task-Diagnostic

[1 paragraph: Reshaping the 3D tensor into 4D by separating reader projections (writers × reader_heads × projection × factors) reveals whether Q, K, and V carry distinct factor subpopulations. On IOI, the three projection modes are >99.9% pure: each mode loads on exactly one of Q, K, or V, with a small but consistent Q↔K mixing coefficient (0.020). On SVA, Q and K *do not separate*: the projection modes mix Q and K at 0.85/0.53. This is structurally informative: IOI uses distinct factor subpopulations for attention-pattern (Q/K) vs value-routing (V) because the circuit has separate roles for "where to look" vs "what to match." SVA's Q and K are coupled because subject-verb agreement works through the joint attention score $QK^\top$ — the same factors help Q find number-bearing subjects and K signal number information. The Q/K mixing pattern is diagnostic of circuit type: identity matrix = distinct-role circuit; Q/K merged = joint-attention circuit. (Appendix I compares 3D, 4D, and 6D decompositions.)]

5.2.10 Scoring vs. Interpretability

[1 paragraph: PCA and Tucker serve complementary roles that happen to converge on the same decomposition. As a *scoring method*, PCA at auto-k is universal and requires no tuning; Tucker at fixed low ranks can beat PCA on specific tasks (IOI: 0.013 vs 0.021) but requires per-task configuration and fails catastrophically with adaptive ranks. As an *interpretability tool*, both methods decompose circuits into the same sub-circuits (cosine 1.000) — PCA is simpler with one hyperparameter k vs three ranks. The principled recommendation: use PCA for both scoring and interpretability; cite Tucker equivalence as validation that the discovered sub-circuits are method-independent.]

6 Experimental Validation

[1 paragraph: We validate the factorization and per-factor attribution with four lines of evidence: (1) null controls (§6.1), (2) metric robustness (§6.2), (3) circuit faithfulness comparison (§6.3), and (4) split-half stability (§6.4). All experiments use GPT-2 Small with $N_F = 8192$ (DST selector, checkpoint atomic-sweep-40).]

6.1 Null Controls

[1 paragraph: Three null baselines. (i) Random 8D subspaces in $\mathbb{R}^{8192}$ (20 trials): project scores$_W$, evaluate CMD. Should be $\gg$0.052 (worse than using all factors). (ii) Shuffled factors (20 trials): permute factor columns of scores$_W$, run PCA $k = 8$, evaluate CMD. Should be $\approx$0.052 (PCA on destroyed structure gives no benefit). (iii) Full-factor baseline: CMD = 0.052. PCA's 0.023 is significant if it beats all controls by a wide margin.]

6.2 Metric Robustness

[FIGURE 6 — NEED: Grassmannian angle heatmap, 5 metrics. logit_diff vs prob_diff = 27° (aligned), entropy vs logit_diff = 82° (orthogonal).]

[1 paragraph: PCA computed independently with five EAP-IG metrics (logit_diff, prob_diff, gated_logit_diff, log_prob, entropy). Finding: the IO-specific metrics (logit_diff, prob_diff) share a common subspace (mean angle 27°) and both yield strong CMD improvement (2.3×, 2.1×). Global metrics (entropy) produce nearly orthogonal subspaces (82°) and no CMD improvement. This is the correct behavior: the PCA subspace reflects *what the metric optimizes*, not an artifact of the factorization. The subspace is stable across metrics that measure the same causal distinction.]

[TABLE 3: Per-metric results. 5 rows (one per metric): variance explained at $k = 8$, full CMD, PCA $k = 8$ CMD, improvement ratio.]

6.3 Circuit Faithfulness

[FIGURE 7 — NEED: Faithfulness curves. Standard EAP vs factorized EAP (full) vs PCA $k = 8$ vs Tucker (10,10,8). x-axis: proportion of circuit kept. y-axis: normalized faithfulness.]

[1 paragraph: Standard MIB faithfulness evaluation. Compare methods at all circuit sizes $k \in [0.001, 1]$: (i) standard EAP (edges ranked by $|\Delta L_{uv}|$), (ii) factorized

EAP with all factors (sub-edges ranked by $|\Delta L(u, i, v, j)|$), (iii) PCA $k = 8$ projected factorized EAP, (iv) Tucker (10,10,8). Report CMD and CPR for each. The key comparison is (ii) vs (iii): PCA projection improves CMD by 2.3×, confirming the denoising effect.]

6.4 Split-Half Stability

[1 paragraph: PCA computed independently on two disjoint halves of the IOI prompts (100 each). Principal angles between the half-subspaces measure stability. Report mean angle at $k = 4$, $k = 8$, $k = 16$. Expected: <15° for top directions confirms the subspace is a property of the circuit, not prompt-specific overfitting. Also compare each half to the full-prompt PCA.]

7 Attribution and Causation Are Orthogonal in Factor Space

7.1 The Threeway Subspace Comparison

[Bolded opener: The factor subspace that explains the most attribution variance is nearly orthogonal to the subspace where causal interventions succeed — ∼88° across 5 tasks.]

[1 paragraph: PCA, Tucker, and DAS each identify a subspace of the 8192-d factor bank. The first two maximize explained variance in the attribution tensor; the third maximizes causal interchange accuracy. We compare all three pairwise via principal angles on the Grassmannian:]

Pair	Mean cosine	Mean angle
PCA vs Tucker	1.000	0°
PCA vs DAS	∼0.03	∼88°
Tucker vs DAS	∼0.03	∼88°

[1 paragraph: PCA and Tucker agree perfectly (as established in §5.2) but both are ∼88° from DAS across IOI, SVA, GT, Gender, Capital. The factors explaining the most variance in "how information flows" are not the factors that matter for "what the model computes." This is not a failure of PCA or Tucker — it is a geometric fact about the relationship between attribution structure and causal structure in factor space.]

7.2 The Activation-Causal Gap

[Bolded opener: Factors activate at $r = 0.97$–0.99 correlation across tasks but have causal effects at $r = 0.03$–0.43, a 14–40× gap.]

[1 paragraph: For each factor, compute activation profile (how strongly it fires across tasks) and causal profile (how much causal effect per task). Activation profiles are nearly identical across tasks ($r \geq 0.97$): every factor fires the same regardless of task. Causal profiles are highly task-specific (r as low as 0.03): which factors *matter* varies dramatically. This gap is invisible to scalar edge scores and explains why the attribution subspace (dominated by high-activation factors) is orthogonal to the causal subspace (dominated by task-specific factors).]

7.3 Background/Modulation Decomposition

[Bolded opener: Tucker C modes capture the modulation component — necessary for task performance but not sufficient in isolation.]

[1 paragraph: Activation projection experiments using `factor_acts_enabled()` hooks. Project factor activations $h = x \cdot F^{\top}$ onto/orthogonal-to Tucker C subspace at runtime:]

k modes	Sufficiency (keep C only)	Necessity (remove C)
1	−30.4% (reversed)	99.6%
2	−29.3% (reversed)	97.7%
4	−15.8% (reversed)	94.3%
8	−16.8% (reversed)	78.9%

[1 paragraph: Removing 8 Tucker C modes drops logit diff by 21% (from 0.1956 to 0.1544) — necessary. But projecting to *only* Tucker modes reverses the logit diff (negative values = model predicts the wrong name) — not sufficient. Like keeping only the AC component of a signal and discarding the DC offset. This resolves the activation-causal gap: factors activate universally because the background dominates activation patterns (baseline computation sustaining coherent predictions). Causal effects are concentrated because only the modulation component carries task-specific information. The 14–40× gap is the ratio of background to modulation magnitude.]

[1 paragraph: Note on the factor bank projection null: projecting the factor bank F (not activations) onto the Tucker C subspace produces zero effect at all ranks (k=1,2,4,8). This is expected, not a failure: F is (8192, 768), so any rank-8 subspace in factor-index space maps to a full-rank-768 output space. The intervention is too weak at the weight level. The correct intervention is activation projection (above), which constrains how factors are *read* during forward computation. Details in Appendix S.]

7.4 Per-Counterfactual Tucker Stability

[Bolded opener: Tucker modes are partially stable across IOI counterfactual types — not identical, not orthogonal — indicating the structure captures a mix of model-intrinsic geometry and corruption-specific routing.]

[1 paragraph: EAP-IG computed separately per IOI counterfactual type (s2_io_flip, s1_io_flip, abc, compound). Tucker (10,10,8) decomposed independently for each. Variance explained varies: s1_io_flip is much more compressible (83.3%) than compound (39.5%). Cross-counterfactual C mode comparison:]

Pair	Matched cos	Mean angle	Factor Jaccard
s2_io vs compound	0.440	58.0°	0.123
s2_io vs abc	0.415	60.6°	0.139
abc vs compound	0.242	73.7°	0.072
s1_io vs abc	0.240	74.1°	0.088
s1_io vs compound	0.191	78.0°	0.046
s2_io vs s1_io	0.145	78.4°	0.111

[1 paragraph: s2_io_flip and compound share the most structure (5/8 modes with cosine >0.64, then 3 nearly orthogonal) — expected since compound includes s2_io_flip. s1_io_flip is most different from all others (angles ∼78°), consistent with its simpler

attribution structure. Factor overlap is low across all pairs (Jaccard 0.05–0.14) — different corruptions route through different factors even when mode directions partially agree. Full tables in Appendix L.]

7.5 Factor-Level Discrimination

[FIGURE — HAVE: Factor-level discrimination analysis showing separation between circuit and non-circuit factor usage. Confirms that circuit-relevant factors form a distinguishable population. See fig2_discrimination.png in BATCH2/.]

[1 paragraph: Factor-level discrimination analysis confirms that circuit-relevant factors are statistically distinguishable from background factors. The separation is consistent with the factor overlap results (§5.2.5): circuit edges use sparse, disjoint factor subsets, creating clear signal above the background distribution.]

7.6 Cross-Task Factor Orthogonality

[Bolded opener: Different tasks use nearly orthogonal factor subspaces: mean cosine of principal angles between top-8 PCA subspaces is 0.03–0.11 across all 10 task pairs.]

[1 paragraph: For each of 5 tasks (IOI, GT, SVA, Gender Bias, Capital-Country), we compute the top-8 PCA subspace of the (writers $\times$ readers $\times$ factors) attribution tensor. Pairwise subspace overlap (mean cosine of principal angles) is uniformly low: GT–SVA and GT–Gender are 0.03, the highest pair (IOI–Capital) is 0.11. This is not an artifact of factor sparsity: random 8D subspaces of $\mathbb{R}^{8192}$ would yield cosine $\sim$0.001. The observed 0.03–0.11 is above random but far below alignment — each task's circuit occupies its own region of factor space.]

[FIGURE — HAVE: 5$\times$5 subspace overlap heatmap (IOI, GT, SVA, Gender, Capital). Diagonal = 1.0, off-diagonal 0.03–0.11. RdBu colormap. See subspace_overlap.png in candidates/.]

[1 paragraph: Tucker mode specificity confirms this at the component level. For each task, the best Tucker C mode concentrates 1.5–7.7% of circuit energy in its own task's circuit but only 0.3–0.8% in other tasks' circuits. SVA modes are the most task-specific (7.7% own vs 0.3–0.4% others); IOI modes are the least specific (1.5% own vs 0.4–0.7% others), consistent with IOI's higher circuit complexity requiring more distributed factor usage. The specificity matrix is strongly diagonal-dominant: each task's modes explain its own circuit, not others.]

[FIGURE — HAVE: 5$\times$5 mode specificity heatmap showing % circuit energy in best mode. Diagonal dominant (1.5–7.7%) vs off-diagonal (0.3–0.8%). See mode_specificity.png in candidates/.]

7.7 Within-Task Sub-Variable Separation

[Bolded opener: Orthogonality holds not only across tasks but within a single task: IOI sub-variables (IO name, Subject, ABC, Random, Full flip) route through nearly orthogonal factor subspaces.]

[1 paragraph: Training separate DAS directions per IOI counterfactual type at Layer 5 ($k = 4$) reveals that the learned causal subspaces are near-orthogonal. Grassmannian distances between all 10 subtask pairs range from 0.60 (IO name vs Full flip) to 2.12 (IO name vs Subject), with 8 of 10 pairs exceeding $\pi/2 \approx 1.57$ (orthogonal). IO

name and Full flip are the closest pair (0.60 radians), which is expected since Full flip includes the IO name swap. All other pairs are at or beyond orthogonality.]

[FIGURE — HAVE: 5×5 Grassmann distance heatmap. IO flip–Full flip = 0.60, all others 1.75–2.12. $\pi/2 \approx 1.57$ **= orthogonal; values** $> \pi/2$ **= beyond orthogonal. See grassmann_heatmap_GOOD.png in candidates/.]**

[1 paragraph: The factor-level picture confirms this. Each subtask's DAS direction selects ∼17–20 private factors (out of its top 20), with only 1.0 factor shared on average per pair. MDS embedding of Jaccard distances shows five well-separated clusters: IO name and Full flip share 9 factors (thick edge), all other pairs share 0–4. Out of 82 unique factors identified by DAS across all 5 subtasks, 68 are private to a single subtask and only 14 are shared by 2 or more. The factor bank provides the vocabulary; each sub-variable selects its own words.]

[FIGURE — HAVE: MDS of Jaccard distance between IOI subtask factor sets (DAS top-20, Layer 5). Five colored nodes with edge thickness $\propto$ **shared factors. IO name–Full flip = 9 shared; most pairs = 0–1 shared. Chance expects <1 shared. See 12a_jaccard_mds_v2.png in candidates/.]**

[1 paragraph: Constrained PCA (unsupervised) and DAS (supervised) find almost entirely disjoint factor sets: of 20 factors selected per subtask, mean overlap is 1.0 factor. DAS uses 82 unique factors across subtasks while CPCA uses only 27 — the supervised method explores a much larger region of factor space because it can find subtle causal directions that don't dominate variance. This reinforces the attribution–causation orthogonality (§7): the factors that explain the most variance (CPCA) are not the factors that carry causal signal (DAS), and this holds per-subtask, not just globally.]

[FIGURE — HAVE: Three-panel figure. Left: DAS factor overlap matrix (5×5, 82 unique factors). Center: CPCA factor overlap matrix (5×5, 27 unique factors). Right: private factors per subtask (DAS ∼17 vs CPCA ∼2). Mean overlap 1.0/20 between methods. See pca_vs_das_overlap.png in candidates/.]

8 Related Work

Sparse autoencoders and activation-space decomposition. [3-4 sentences: SAEs learn interpretable directions in activation space. Our factor bank is analogous but in weight space: the model computes through factors, so they are guaranteed to participate in computation. SAEs decompose what the model outputs; factors decompose how it computes. Cite Bricken et al. 2023, Huben et al. 2023, transcoders. Factorized DAS with group lasso identifies 9 active factors for IOI — but individual factors are gauge-dependent; the causally meaningful object is the subspace they span.]

SAE features and factors span the same subspace but use different bases. Projecting 24,576 JumpReLU SAE features (Bloom et al., layer 8, GPT-2 Small) onto the 8,192-factor bank recovers 99.8% of variance on average, yet individual alignment is low: the maximum cosine similarity between any SAE feature and its nearest factor is 0.37, with a mean of 0.15 across all features. This holds regardless of which layer's SAE is used (layers 5, 7, 8, 9 all yield max cosine $\leq$**0.34 with the top circuit factors). The two decompositions are complementary: SAE features are monosemantic directions in per-layer activation space, while factors are shared communication channels in weight space. Critically, because factors share a global index across all layers, 8% of factors**

active in the IOI circuit participate in two or more pathway stages (top-20 per edge), rising to 17% at top-50 — cross-layer tracking that per-layer SAE dictionaries cannot provide.

Unsupervised subspace decomposition (NDM). [3-4 sentences: Huang & Hahn (2025) learn an orthogonal rotation of the residual stream via neighbor distance minimization, achieving Gini 0.71 on IOI subspace patching. Their PCA baseline fails (Gini 0.38) because PCA on $\mathbb{R}^{d_m}$ activations captures variance, not task structure. Our PCA succeeds (CMD 0.023 vs 0.052 baseline) because the factor-attribution tensor *already encodes task relevance* — PCA denoises rather than discovers. They decompose *where* information is stored; we decompose *how* attribution flows. Their future directions (§6) propose subspace circuits via weight-space analysis — our factorized architecture is exactly this.]

Low-rank structure in weights. [2-3 sentences: Kaushik et al. (2025) show weight matrices across 500+ LoRA adapters converge to a shared low-rank subspace ($\leq$16 principal directions). We find analogous low-rank structure in attribution space within a single model. LoRA's success further supports the low-dimensional weight-space structure we exploit.]

Circuit discovery methods. [2 sentences: ACDC, EAP, EAP-IG, MIB operate at full-edge granularity. Our contribution decomposes each edge into per-factor contributions and finds the optimal low-dimensional basis for those contributions.]

Weight-space circuit discovery. [2-3 sentences: Tower (2026) shows functional roles (name mover, S-inhibition, etc.) are recoverable from weight features alone, without forward passes. Our factorization is complementary: weight features classify *what role* a head plays; per-factor attribution decomposes *what information* flows through each edge. Together, they provide role identification + sub-edge decomposition entirely from weight structure.]

Causal abstraction and DAS. [2-3 sentences: Geiger et al. (2023, 2024). We show the factor manifold provides effective initialization for DAS. Attribution-based subspaces are $\sim$88° from the causal subspace — a geometric finding, not a method improvement. The orthogonality is consistent with Geiger et al.'s observation that causal structure requires interventionist methods, not just observational analysis.]

Conceptor steering and subspace methods. [2-3 sentences: Conceptor matrices (NeurIPS 2024) project activations onto learned subspaces for steering. Factor manifold is a weight-space analogue. Boolean operations (AND/OR/NOT) on factor-identified task subspaces produce valid causal interventions (97–100% IIA). See Appendix P.]

Composition scores. [2-3 sentences: Merullo et al. (2024) "Talking Heads": per-pair SVD of composition. Our shared factor bank provides a unified cross-edge basis. Factorized composition via $S_u^\top(FF^\top)S_v$ is algebraically equivalent to standard composition through the shared factor bank. See Appendix Q.]

9 Discussion

[1 paragraph: We adopt the view that sub-edge circuit structure is captured by factor subspaces — the spans of factors that circuit projections selectively activate — rather than by individual weight matrix ranks or individual factors. The factor bank makes this precise: PCA on the per-factor attribution tensor decomposes circuits into interpretable sub-circuits (IOI: early heads $\rightarrow$ MLP0, S-inhibition fan-out, name movers $\rightarrow$ logits), and Tucker decomposition independently recovers identical factor directions (cosine 1.000, Grassmannian geodesic ≈ 0). This convergence of two mathematically different methods — PCA maximizes variance of the flattened matrix, Tucker minimizes reconstruction error of the 3D tensor — is strong evidence that the sub-circuit structure is real, not an artifact of the decomposition method. Individual factors are basis vectors in the overcomplete dictionary, analogous to SAE encoder directions — interpretable as a catalog, but gauge-dependent; only their spans are mechanistically invariant. The factor bank also provides a geometric inductive bias for causal variable discovery: it defines a low-dimensional submanifold of the Grassmannian where spectral methods succeed (constrained PCA at 98.8% IIA, zero training) and where DAS optimization converges faster (91.2% vs 73.5% on hard examples). The constraint gradient — more geometric constraint yields worse IIA (Riemannian DAS 64% vs vanilla 99.3%) — shows the manifold's value is as initialization, not ongoing constraint.]

[1 paragraph (NEW v6): The $\sim$88° orthogonality between attribution and causal subspaces reveals a fundamental distinction. Attribution captures sensitivity: the directions along which the model's output changes most when the input is perturbed. Causation captures computation: the directions through which the model routes task-relevant information. These are different questions with orthogonal answers in factor space. The background/modulation decomposition makes this concrete: factors activate universally to sustain baseline computation (background), but only a small modulation component carries task-specific signal. Tucker decomposes the modulation component, which is necessary (21% logit diff drop when removed) but not sufficient alone (reversed logit diff when isolated). The 14–40$\times$ activation-causal gap is the ratio of background to modulation magnitude.]

[1 paragraph (NEW v6): Each circuit edge uses a distinct factor subset (cross-pathway Jaccard $< 0.5\%$), and different counterfactual types route through partially distinct factor subspaces (principal angles 58–78°). This suggests the factor bank functions as a high-dimensional routing table: the bank provides the vocabulary, the selectors choose the words, and different edges "say different things" through the same vocabulary. The sparsity of factor usage per edge ($\sim$20 of 8192) enables this multiplexing without interference.]

[1 paragraph: Limitations. (1) The factorized model is an approximation of the pretrained model, so sub-edge structure exists in the factorized representation, not directly in the original weights. Model equivalence (§3.3) mitigates but does not eliminate this concern. (2) GPT-2 Small only. Cross-architecture validation (Qwen, Pythia) is planned but not yet run. (3) Factorization requires training ($\sim$2 GPU-hours for GPT-2 Small); per-factor attribution is then free. (4) Individual factors are gauge-dependent (any rotation of the bank is equally valid). Only factor *subspaces* (PCA/Tucker mode spans) are gauge-invariant. The factorization provides the right overcomplete dictionary to search for subspaces; individual factors are basis vectors, not interpretable units. This parallels how PCA components aren't meaningful indi-

vidually but their span is well-defined. (5) (NEW v6) Hard-IIA results are on $\sim$34 evaluation examples — enough to demonstrate the gradient attractor effect but larger-scale validation is needed. (6) (NEW v6) The background/modulation decomposition is demonstrated on IOI only. Extension to other tasks would strengthen the claim. (7) (NEW v6) Sub-circuit decomposition recovers known structure (validation) rather than discovering new biological insights — the method is the contribution, the IOI findings are validation.]

[1 paragraph: Future directions. (i) Cross-architecture validation: train factorized Qwen-0.5B, Pythia-160M, and test whether Tucker structure (core tensor sparsity, mode interpretability) transfers. (ii) Larger models (GPT-2 Medium/Large). (iii) Integration with weight-space circuit discovery for end-to-end role identification + sub-edge decomposition. (iv) Sparser checkpoints: train factorizations with stronger DST/JumpReLU penalties to achieve $\sim$500–2000 active factors per head, enabling pathway-specific factor identification and single-factor knockout experiments. (v) RAVEL disentanglement: test whether factor subspaces separate entangled properties (e.g., entity vs. attribute). (vi) Applications: circuit-guided fine-tuning (targeting the active factor subspace via FactorLoRA), dynamic inference (gating inactive factors based on selector predictions), conceptor-style compositional steering via factor subspace algebra (AND/OR/NOT).]

Appendix

A Gauge Freedom and QK/OV Distillation

A.1 Rotational gauge freedom in attention

[State the problem: for any orthogonal $R \in O(d_h)$, replacing $W_Q \to W_Q R$ and $W_K \to W_K R$ leaves the QK circuit invariant: $(W_Q R)(W_K R)^\top = W_Q R R^\top W_K^\top = W_Q W_K^\top$. So individual weight matrices W_Q, W_K are not uniquely determined by the model's behavior — only the composite $\mathbf{QK} = W_Q W_K^\top$ and $\mathbf{OV} = W_V W_O$ are gauge-invariant observables.]

[1 paragraph: This means distilling on individual weights ($\|S_Q F - W_Q\|^2$) is ill-posed: there are infinitely many valid targets related by arbitrary rotations R, and the factorization would need to "guess" the right gauge. Distilling on the composite circuits ($\|QK^{\text{fac}} - QK^{\text{tgt}}\|^2$) eliminates this ambiguity entirely. The factorization is free to choose any internal gauge for its S_Q, S_K as long as their product $S_Q F (S_K F)^\top$ matches the target QK.]

A.2 Bias reparameterization

[State the problem: if we zero b_Q, b_K for the factorized model (since factors should capture bias effects), the attention score changes because the cross-terms $W_Q b_K$, $b_Q W_K^\top$, and $b_Q \cdot b_K$ are gauge-dependent and nonzero in the pretrained model.]

[1 paragraph: Solution: compute these cross-terms from the *pretrained* weights before training, store them as frozen buffers (`_qk_bias_q`, `_qk_bias_k`, `_qk_bias_const`), and add them back during attention score computation. This makes the factorized attention scores match the pretrained model exactly even with $b_Q = b_K = 0$. Similarly, b_V is folded through W_O into b_O. See `reparameterize_layer_biases_` in the codebase.]

B EAP Derivation

B.1 Standard EAP

[COPY FROM Ivan's draft §2.1 (overleaf_draft.tex lines 196–292). The full derivation from $r_u^{(n)} = z_u^{(n)} W_u^{o\top}$ and $\text{act}_v^{(n)} = \text{LN}(\text{resid}_v^{(n)}) W_v^{in}$ to the EAP formula with shape annotations. Already publication-ready.]

B.2 Factorized EAP

[COPY FROM Ivan's draft §2.2 (overleaf_draft.tex lines 404–512). The four-line algebraic expansion substituting $W = SF$ into the EAP formula, arriving at $\Delta L_{uv} = \sum_{i,j} \Delta L(u, i, v, j)$. Already publication-ready. Includes the intuitive description: "$\Delta L(u, i, v, j)$ answers: how much does writing by module u into factor subspace i, as read by module v through factor subspace j, affect the loss?"]

B.3 EAP vs. EAP-IG

[1–2 paragraphs:

Paragraph 1 (decomposition is exact for both): We present the factored decomposition (Proposition 1) for first-order EAP. The decomposition extends immediately to EAP-IG by linearity: at each integration step s the substitution $W = SF$ is the same algebra, so the per-factor score at step s sums to the edge score at step s. Averaging over steps preserves the equality: $\Delta L^{\mathrm{IG}}_{uv} = \sum_{i,j} \Delta L^{\mathrm{IG}}(u,i,v,j)$ where each per-factor EAP-IG score is simply the average of per-factor scores across integration steps. No new proof is required.

Paragraph 2 (closed-form shortcut is EAP-only): What *does not* extend to EAP-IG is the closed-form shortcut (Appendix C), which pre-computes the LayerNorm Jacobian as a single matrix M. Under EAP-IG the Jacobian varies at each interpolation point, so M would need to be recomputed per step. Our implementation handles this correctly by running the factored decomposition at each IG step independently via `attribute_rotation.get_scores_eap_ig`. All scores reported in the main text use factorized EAP-IG (5 integration steps), which computes exact per-factor scores at each step and averages.]

C Closed-Form Factorized EAP Score

C.1 LayerNorm Jacobian

[COPY FROM Ivan's draft Appendix A (overleaf_draft.tex lines 590–751). Full derivation of $J_{\mathbf{LN}} = \frac{1}{\sigma}(I - \frac{1}{d_m}\mathbf{1}\mathbf{1}^\top - \frac{1}{d_m}\hat{x}\hat{x}^\top)$. Already publication-ready with all intermediate steps.]

C.2 Expanding $F_i J^{(n)\top}_{\mathbf{LN},v} F_j^\top$

[COPY FROM Ivan's draft Appendix B §B.1 (lines 757–829). Introduction of the Gram matrix $M_{ij} = F_i F_j^\top - \frac{1}{d_m}(F_i\mathbf{1})(\mathbf{1}^\top F_j^\top)$ and the factor encoding $r_v^{(n)} = F\hat{x}_v^{(n)}$. Arrives at the scalar form of the link term.]

C.3 Final closed-form score

[COPY FROM Ivan's draft Appendix B §B.2–B.3 (lines 831–974). The writer scalar $A^{(n)}_{u,i}$, reader scalar $C^{(n)}_{v,j}$, substitution chain arriving at $\Delta L(u,i,v,j) = M_{ij}T_1^{(uv)} - \frac{1}{d_m}T_2^{(uv)}$. Einsum forms for T_1 and T_2. Cache contents table.]

D Proofs of Theorems 1–3

D.1 Theorem 1: Eckart-Young-Mirsky (Optimal Linear Compression)

[State the theorem formally for our setting. Let $S \in \mathbb{R}^{E\times N_F}$ be the attribution matrix with SVD $S = U\Sigma V^\top$. For any rank-k matrix B, $\|S - U_k\Sigma_k V_k^\top\|_F \le \|S - B\|_F$. Apply: at $k = 8$, reconstruction error = 40.8% of total variance, which is the minimum achievable by any rank-8 projection.]

D.2 Theorem 2: Davis-Kahan Subspace Stability

[State the $\sin\Theta$ theorem. If $S = S_{\textbf{true}} + E$ and $\delta_k = \sigma_k - \sigma_{k+1}$ is the singular value gap, then $\sin\angle(\hat{V}_k, V_k) \leq \|E\|_{\textbf{op}}/\delta_k$.

Estimate noise level: the Marchenko-Pastur threshold for an $E \times N_F$ random matrix gives $\sigma_{\textbf{noise}} = 0.0002$.

Compute bounds:

- **$k = 4$: $\delta_4 = 0.0023$, bound = $0.0002/0.0023 = 0.087$, $\arcsin(0.087) \approx 5.0$.**
- **$k = 8$: $\delta_8 = 0.0003$, bound = $0.0002/0.0003 = 0.667$, $\arcsin(0.667) \approx 41.8$.**
- **$k = 16$: $\delta_{16} \approx 0$, bound $\to \infty$, angle $\to 90$.**

]

D.3 Theorem 3: Active Subspace Error Bound / SURE Rank Selection

[If using active subspace (Constantine 2015): State the Poincaré-type inequality. The MSE from projecting the metric onto the top-k subspace is bounded by $\sum_{i>k} \lambda_i/(\lambda_k - \lambda_{k+1})$. Compute: $k = 4$ bound = 14.2, $k = 8$ bound = 144.1. The 10× gap reflects the primary/secondary eigenvalue separation.]

[If SURE results available: State Stein's lemma for the truncated SVD denoiser. The unbiased risk estimate SURE$(k) = \|S - S_k\|_F^2 + 2\text{df}(k)\hat{\sigma}^2$. The divergence term df$(k)$ for truncated SVD counts effective degrees of freedom. Plot SURE(k) for $k = 1 \dots 64$, show minimum at $k \approx$?. This provides non-circular rank selection.]

E Training Details

E.1 Optimization

[Optimizer: Adam. Learning rate schedule: reactive halving — if $\mathcal{L}_{\textbf{total}} > 2 \times \mathcal{L}_{\textbf{prev}}$, skip the step and halve LR (bounded below by `min_lr`). No warmup, no cosine. Only F and S parameters are optimized (deduplicated by `id()`); all other parameters frozen from pretrained model.]

E.2 Loss components

[Core loss: $\mathcal{L}_{\textbf{core}} = \sum_i \|QK_i^{\textbf{fac}} - QK_i^{\textbf{tgt}}\|^2 + \|OV_i^{\textbf{fac}} - OV_i^{\textbf{tgt}}\|^2$, summed over layers.

Auxiliary loss: selector-specific sparsity penalty.

- **L1: $\lambda_{\textbf{aux}} \cdot$ mean$(|S|)$**
- **JumpReLU: $\lambda_{\textbf{aux}} \cdot L_0(S)$ (straight-through estimator)**
- **DST: $\lambda_{\textbf{aux}} \cdot |\theta|$ (learnable per-channel threshold)**

Optional: entropy loss (within-head + across-model deficit), factor contrastive loss (pairwise $\exp(|\cos| \cdot \text{mult}) - 1$ on learned factors).]

E.3 Selector types

[TABLE A1: Selector types. Type / Mechanism / Auxiliary loss formula / Masking. L1: soft penalty, continuous. JumpReLU: per-channel threshold, hard (STE). JumpReLU-tanh: JumpReLU + tanh-scaled magnitude, hard (STE). DST: per-channel learnable threshold + BinaryStep STE, hard (STE).]

E.4 Hyperparameter ranges

[$N_F \in \{1024, 2048, 4096, 8192, 16384, 32768, 40000\}$. $\lambda_{\textbf{aux}}$ swept per selector type. At $N_F = 32768$: ~100 active factors per channel (~0.3%). Pareto sweep: trained >200 checkpoints across bank sizes and selector types. Frontier extracted by non-dominated sorting on (CE, L0/channel).]

[FIGURE A1: Pareto frontier by bank size, all checkpoints.]

E.5 Model Equivalence Details

[CREH evaluation methodology. MAS IIA computation. CE gap measurement. Bidirectional DAS interchange protocol. Details for both GPT-2 and Qwen.]

F Extended Results

F.1 Full CMD/CPR tables

[CMD and CPR at $k = 1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 8192$ for PCA, plus full baseline and DAS $k = 1, 2, 4, 8$.]

F.2 Per-checkpoint faithfulness curves

[Faithfulness curves for all checkpoints shown in the Pareto analysis, separated by selector type. Show that the PCA improvement is not checkpoint-specific.]

F.3 Split-half raw data

[Full principal angle vectors between half-subspaces at $k = 4, 8, 16$. Comparison of each half to the full-prompt PCA.]

F.4 Null control distributions

[Histograms of CMD for (i) 20 random 8D subspaces, (ii) 20 shuffled-factor PCA runs. Mark PCA $k = 8$ CMD and full baseline CMD on the histogram.]

F.5 SURE risk curve

[If computed: SURE(k) for $k = 1 \ldots 64$. Mark the minimum.]

F.6 Variance explained spectrum

[Scree plot: singular values $\sigma_1 \ldots \sigma_{64}$ and cumulative variance explained. Mark the primary/secondary gap at $k = 4$ and the secondary/tail transition at $k = 8$.]

F.7 Cross-Architecture Results

[DEFERRED to future revision. Cross-architecture validation planned for Qwen-0.5B and Pythia-160M.]

G Alternative Decomposition Methods

[TABLE A2: Full method comparison.]

[1 paragraph: We evaluated decomposition methods on the factorized attribution tensor to verify that PCA/Varimax's performance is not an artifact of method choice. Table A2 reports CMD for all methods at their best k. PCA and Varimax dominate; ICA (statistical independence) and PLS-DA (supervised labels) both underperform unsupervised PCA, confirming that the low-rank linear structure in the attribution tensor is the primary signal, and that rotating within the PCA subspace (Varimax) extracts additional benefit from the sparsity of factor selectors.]

G.1 ICA

[2-3 sentences: FastICA at $k = 2, 4, 8, 16$. Best result at $k = 16$ (CMD 0.032), worse than PCA at $k = 8$ (0.023). ICA seeks statistically independent components, which is a weaker criterion than variance maximization for this problem — the attribution signal is low-rank and approximately Gaussian in the dominant dimensions, so PCA is better suited.]

G.2 PLS-DA

[2-3 sentences: Partial Least Squares Discriminant Analysis using Wang et al. (2023) IOI circuit labels (13,795 circuit edges). Best result at $n = 8$ components (CMD 0.045), better than the full baseline but worse than unsupervised PCA. The supervised labels help concentrate signal but PLS overcommits to the binary circuit/background separation at the expense of fine-grained within-circuit ranking. The unsupervised factor structure is richer than the binary labels.]

H Stability and Enrichment Analysis

H.1 Bootstrap Stability of Principal Components

[Bolded opener: The leading PCA components are unstable under bootstrap resampling, while tail components are rock-solid — inverting the normal expectation and explaining the U-shaped CMD curve.]

[1 paragraph: 50 bootstrap PCA runs (resampling edges with replacement). Principal angles between each bootstrap subspace and the full-data PCA subspace: PC0 at 89° (essentially random), PC1 at 85°, PC2 at 76°. But PC6 at 2.7°, PC7 at 0.2°. No individual factor achieves stability > 0.5 (the highest is f44 at 0.32). Interpretation: the dominant variance (PC0–2) is driven by outlier edges — high-variance positional/embedding structure that changes with the edge sample. The robust structure lives in the smaller components (PC6–7), which capture the stable computational pathway (induction $\rightarrow$ S-inhibition $\rightarrow$ name mover). This explains why $k = 8$ works: it

captures enough stable structure while the leading-component noise is diluted by the more stable tail components in the norm-based ranking.]

[FIGURE A2 — NEED: Principal angle vs. component index for 50 bootstrap runs. Error bars showing spread. Clear two-tier structure: PC0-3 at 50–90°, PC4-7 at 0–15°.]

H.2 GSEA Enrichment of Principal Components

[Bolded opener: Gene Set Enrichment Analysis on factor loadings reveals that duplicate token heads cluster in PC0–1 (6× enrichment, $p = 0.001$) while induction heads cluster in PC3/PC6.]

[1 paragraph: We treat each PC's factor loadings as a ranked list and test enrichment for Wang et al. head categories (duplicate token, induction, S-inhibition, name mover, backup name mover) using standard GSEA permutation tests. Duplicate token heads show strong enrichment in PC0–1 (top-100 factors: 6× enrichment, $p = 0.001$). Induction heads cluster in PC3 and PC6. Name movers do not show significant enrichment in any single PC — they likely use distributed factor patterns that span multiple components. This categorical structure in the principal components provides interpretability: PCs are not arbitrary directions but correspond to functional head categories in the IOI circuit.]

H.3 Convergent Factors Across Methods

[1 paragraph: Three factors appear consistently across PLS-DA and stability analysis: f44 (highest bootstrap stability at 0.32), f1798, and f4461. These are candidates for "core IOI factors" — directions in the factor bank that are both computationally important (high PCA loading) and structurally robust (high bootstrap stability). However, individual factor identity is gauge-dependent (§8), so these should be interpreted as markers of a robust subspace, not as individually interpretable features.]

I Tucker Dimensionality Comparison

[1 paragraph: We compare three reshapings of the attribution tensor for Tucker decomposition: 3D (writers × readers × factors, standard), 4D (writers × reader_heads × projection × factors), and 6D (wr_layer × wr_head × rd_layer × rd_head × projection × factors). All use HOSVD with randomized truncated SVD on checkpoint atomic-sweep-40, IOI task.]

[TABLE A1: Variance explained by decomposition.]

[1 paragraph: 3D captures the most variance (59.75%) because writer/reader modes can freely mix layer/head/projection structure. 4D and 6D impose structural constraints that reduce total variance capture but yield more interpretable modes. For CMD scoring, 3D is the default (Section 5.2). The 4D/6D decompositions are diagnostic tools: they reveal whether the circuit has distinct Q/K/V subpopulations.]

I.1 Factor Subspace Equivalence

[TABLE A2: Principal angles between factor mode subspaces.]

[1 paragraph: 4D and 6D factor modes span the same subspace (mean angle 0.2°, max 0.7°): separating writer layers from heads does not change which factors matter. 3D factor modes differ substantially (68.3°) because the flat writer/reader axes force structural information into the factor modes. The 3D decomposition conflates structure and content; 4D/6D separate them. For applications that need the factor subspace specifically (e.g., ablation experiments), the 4D decomposition is preferable. For circuit scoring, the 3D decomposition is preferable because it captures more total variance.]

I.2 Q/K/V Projection Modes by Task

[TABLE A3: Projection mode purity by task.]

[1 paragraph: On IOI, projection modes separate Q, K, and V with >99.9% purity (Table A3). A small but consistent Q↔K mixing coefficient (0.020) reflects the attention-score duality $QK^\top$. On SVA, V separates cleanly but Q and K mix at 0.85/0.53 — the circuit uses Q and K jointly for number agreement, so the attribution distributes symmetrically across both projections. The 4D variance explained drops from 42.73% (IOI) to 13.98% (SVA) because SVA's signal is MLP-dominated: separating reader projections splits 99% of the signal away from the 1% in attention. The projection separation pattern is *diagnostic of circuit type*: identity matrix = distinct-role circuit (IOI), Q/K merged = joint-attention circuit (SVA).]

I.3 ICA and Varimax

[1 paragraph: We apply ICA (FastICA) and varimax rotation to the 8 factor modes from each decomposition. Neither improves sparsity (Gini shift <0.02). The HOSVD modes are already near-optimal — the useful structure is in *which* 8 of 8192 factor directions to use, not in how to rotate within those 8.]

J Tucker Scoring: Full Results and Ablations

J.1 Scoring Aggregation Methods

[1 paragraph: Two aggregation methods for converting Tucker-reconstructed factor scores into edge scores. abs_of_sum: $|\sum_f \tilde{T}[w,r,f]|$ matches standard EAP-IG's aggregation. sum_of_abs: $\sum_f |\tilde{T}[w,r,f]|$ counts total unsigned factor contribution. The optimal method varies by task.]

Table 4: Scoring aggregation comparison (CMD, lower is better).

Task	Config	sum_of_abs CMD	abs_of_sum CMD
IOI	(10,10,8)	**0.013**	0.028
GT	(20,20,16)	0.034	**0.027**
SVA	(15,15,8)	0.028	0.022

[1 paragraph: sum_of_abs works better for IOI because the circuit has many factors contributing with mixed signs along each edge — absolute-value-then-sum captures total factor involvement. abs_of_sum works better for GT because the dominant factors

are aligned in sign within each edge, so summation preserves signal while cancellation removes noise. Both methods should be tested; we report the best for each task in the main text.]

J.2 Sparse Factor Modes

[1 paragraph: We test thresholding each factor mode $U_f[:, k]$ to its top-m entries by absolute value before Tucker reconstruction. This removes noise factors while preserving task-relevant ones.]

Table 5: Effect of sparse factor modes on Tucker CMD.

Factor sparsity	IOI CMD	GT CMD
Dense (all 8192)	**0.013**	0.027
Sparse-200	0.018	0.018
Sparse-50	0.018	**0.018**
Sparse-20	0.018	0.028

[1 paragraph: For GT, sparse-50 is better than dense (0.018 vs 0.027) — the top 50 factors per mode capture the task-relevant signal and removing the remaining factors eliminates noise. For IOI, dense is marginally better (0.013 vs 0.018) because IOI's richer factor structure benefits from retaining all contributions. The optimal sparsity level correlates with the factor effective rank (Table ?? in §5): higher effective rank → denser modes preferred.]

J.3 HOSVD vs ALS Refinement

[1 paragraph: We compare HOSVD initialization alone (randomized truncated SVD, no iterative refinement) against HOSVD + Alternating Least Squares refinement (100 iterations, tolerance 10^{-5}). ALS refinement is critical for scoring quality.]

Table 6: HOSVD-only vs HOSVD+ALS refinement (CMD, lower is better).

Task	HOSVD-only CMD	HOSVD+ALS CMD	Improvement
IOI	0.170	**0.013**	13×
GT	0.080	**0.018**	4.4×
SVA	0.069	**0.022**	3.1×

[1 paragraph: HOSVD-only initialization is catastrophically worse: 13× worse CMD on IOI, 4.4× on GT, 3.1× on SVA. The randomized SVD provides a coarse starting point, but the alternating least squares steps are essential for accurate reconstruction of the fine-grained factor structure. We use HOSVD+ALS (tensorly's `tucker` with `init="svd"`, `n_iter_max=100`) throughout.]

J.4 Negative Results: Approaches That Did Not Improve Scoring

J.4.1 Asymmetric Tucker Ranks

[1 paragraph: Setting mode ranks independently (e.g., $r_w = 15, r_r = 1, r_f = 8$ for SVA's rank-1 reader structure) produces CMD 0.33–0.81. Even when a mode has effective rank 1, compressing it to rank 1 in Tucker removes the 20% of variance in non-dominant modes that contains essential edge-discrimination signal. The rank-1 degeneration described in §5.1.2 means Tucker *cannot improve rankings* via the dominant mode, but aggressively compressing the non-dominant modes actively *destroys* signal.]

J.4.2 CPCA Pre-filtering

[1 paragraph: We tested restricting the factor dimension to factors identified by constrained PCA (top 1–10% by selector magnitude) before Tucker decomposition. This did not help scoring:]

Table 7: CPCA pre-filtering on SVA (CMD, lower is better).

Method	Factors retained	CMD
Standard EAP-IG (all factors)	8192	**0.015**
CPCA filtered 5% (no Tucker)	409	0.038
CPCA filtered 10% (no Tucker)	819	0.047
Best CPCA + Tucker	409–819	0.182

[1 paragraph: The top 1% of factors by selector magnitude holds only 10% of the EAP-IG attribution energy. The remaining 90% is distributed across "inactive" factors that still carry real attribution signal. CPCA pre-filtering works for DAS (§4.3) because DAS operates at a single layer where the selector identifies the right causal subspace. EAP-IG aggregates across all layers and all edges, so the attribution signal is broadly distributed and pre-filtering removes real signal, not just noise. This is a fundamental difference between causal intervention (DAS) and gradient-based attribution (EAP-IG).]

J.4.3 SVA Edge-Case Methods

[1 paragraph: No Tucker variant, rank configuration, scoring method, or pre-processing step improved on SVA's standard EAP-IG CMD of 0.015. We tested: factor-only PCA ($k = 2$ to 512), soft thresholding (0.5σ to 5σ), top-k factors per edge ($k = 10$ to 1000), and high-reader-rank Tucker ($r_r = 50, 100, 445$). SVA's circuit is structurally incompatible with Tucker denoising — it is already clean, and the rank-1 reader structure means no Tucker configuration can change edge rankings (see §5.1.2). The correct interpretation is that SVA's 0.015 CMD is the floor for this checkpoint and task.]

J.5 Cross-Task Circuit Structure

J.5.1 IOI Sub-Circuit Recovery

[1 paragraph: Tucker (10,10,8) on IOI's circuit sub-tensor (top 1% of edges by standard EAP-IG) recovers the complete Wang et al. (2022) IOI circuit automatically. Each

head loads on a single writer mode with loading > 0.99. The modes are perfectly interpretable:]

Table 8: IOI Tucker sub-circuits (top 8 core tensor entries, accounting for 67% of total energy).

#	Sub-circuit	Energy	Tier	Known IOI Role
1	a0.h1 → m0 via F0	13.8%	Primary	Duplicate token → MLP0
2	a0.h4 → m0 via F1	11.5%	Primary	Early attn → MLP0
3	a0.h5 → m0 via F2	10.6%	Primary	Early attn → MLP0
4	a0.h3 → m0 via F3	9.7%	Primary	Self-attn → MLP0
5	a5.h5 → m5 via F4	7.6%	Secondary	Induction → MLP5
6	a5.h5 → a8.h6⟨v⟩ via F5	5.5%	Secondary	Induction → S-inhibition
7	a5.h5 → m5 via F6	4.5%	Secondary	Induction → MLP5
8	a9.h9 → logits via F7	3.8%	Secondary	Name mover → output

[1 paragraph: Two-tier structure emerges: the *primary tier* (#1–4, 46% of energy) is infrastructure — layer-0 heads routing through MLP0, which every task needs for embedding processing. The *secondary tier* (#5–8, 21%) is the actual IOI algorithm: induction → S-inhibition → name mover, the pathway identified by manual circuit analysis.]

J.5.2 SVA: Shallow Rank-1 Circuit

[1 paragraph: All 8 SVA sub-circuits route through a single reader (m0):]

Table 9: SVA Tucker sub-circuits (top 5 core tensor entries).

#	Sub-circuit	Energy
1	a0.h5 → m0 via F0	29.2%
2	a0.h3 → m0 via F1	16.3%
3	a0.h1 → m0 via F2	14.4%
4	a0.h4 → m0 via F3	12.5%
5–8	a0.h{11,10,6} + input → m0	24.8%

[1 paragraph: No mid-layer or late-layer structure in the top-1% circuit. SVA is a "shallow" task: subject-verb agreement is processed primarily through early attention heads routing to MLP layer 0. The reader rank-1 structure (Table ??) is not just a scoring diagnostic — it reflects the circuit's actual architecture. There is no multi-layer relay; information goes directly from layer-0 attention to MLP0.]

J.5.3 Greater-Than: MLP Cascade

[1 paragraph: GT shares IOI/SVA's primary structure (early heads → m0) but has a distinctive secondary structure — a 3-layer MLP cascade:]

[1 paragraph: Factor modes F6 and F7 are *shared* between the a0.h1→m1, a0.h1→m2, m0→m1, and m0→m2 sub-circuits. This sharing means the same factor subspace carries the "number representation" through consecutive MLP layers for greater-than

Table 10: Greater-Than Tucker sub-circuits (top 10 core tensor entries).

#	**Sub-circuit**	**Energy**	**Role**
1–6	a0.h{1,3,5,4,7,10} $\rightarrow$ m0	82%	Primary: early $\rightarrow$ MLP0
7	a0.h1 $\rightarrow$ m1 via F6	3.8%	MLP cascade layer 1
8	a0.h1 $\rightarrow$ m2 via F7	3.3%	MLP cascade layer 2
9	m0 $\rightarrow$ m1 via F6	0.5%	MLP cascade: m0 $\rightarrow$ m1
10	m0 $\rightarrow$ m2 via F7	0.3%	MLP cascade: m0 $\rightarrow$ m2

comparison. The MLP cascade (m0$\rightarrow$m1$\rightarrow$m2) is GT's distinctive secondary structure, absent in SVA (rank-1 readers) and present in a different form in IOI (attention relay rather than MLP cascade).]

J.5.4 Two-Tier Universal Structure

[1 paragraph: All three tasks share the same *primary structure*: layer-0 attention heads $\rightarrow$ MLP0 through task-distinct factor subspaces (F0–F5). This is infrastructure — embedding processing that every task requires. The primary tier accounts for 46% (IOI), 97% (SVA), and 82% (GT) of circuit energy. Tasks differ in their *secondary structure*: IOI has a multi-layer attention relay (layers 5$\rightarrow$8$\rightarrow$9), GT has an MLP cascade (m0$\rightarrow$m1$\rightarrow$m2), and SVA has no secondary structure at all. The secondary tier is where task-specific computation happens; Tucker separates it from the universal infrastructure via the core tensor's block structure.]

J.6 4D Tucker: Q/K/V Separation

[1 paragraph: Reshaping the reader axis from flat 445 nodes to (157 head-components $\times$ 4 Q/K/V/MLP types) yields a 4D tensor. Tucker on this 4D tensor reveals whether a circuit uses Q, K, and V independently or jointly.]

Table 11: 4D Tucker Q/K/V projection mode purity by task.

Task	**Mode 0**	**Mode 1**	**Mode 2**	**QKV purity**
IOI	Q (1.000)	V (1.000)	K (1.000)	>99.9%
SVA	V (1.000)	K(0.85)/Q(0.53)	Q(0.85)/K(0.53)	Mixed
GT		[TBD — pod pending]		

[1 paragraph: IOI separates Q, K, V with >99.9% purity — each projection type becomes an independent Tucker mode. The small Q$\leftrightarrow$K mixing (0.020) reflects the $QK^\top$ attention score duality. SVA separates V cleanly but mixes Q and K (0.85/0.53 loadings), indicating the circuit uses Q and K jointly for number agreement. This pattern is *diagnostic of circuit type*: identity-like projection modes indicate distinct-role circuits (IOI), while Q/K merging indicates joint-attention circuits (SVA). The 4D variance drops from 42.7% (IOI) to 14.0% (SVA) because SVA is MLP-dominated — separating attention projections isolates a minor fraction of the total signal.]

K Weight-Space Communication Analysis

K.1 Communication Matrices Are Full-Rank

[1 paragraph: For each edge $(u \to v)$, the communication matrix $C_{uv} = W_O^{u\top} W_V^v$ is $(d_{\textbf{head}} \times d_{\textbf{head}})$. We compute singular value concentration $\sigma_1 / \sum_i \sigma_i$ across all IOI edges and random baseline edges. Result: concentration is 3.9–5.5% for all pathway types (previous-token $\to$ induction, induction $\to$ S-inhibition, S-inhibition $\to$ name-mover) and 4.6% for random edges. Communication is distributed across all 64 singular values with no dominant channel. This is why the nuclear norm (sum of all σ_i) is the appropriate summary statistic, and why rank-1 dominance tests capture only ∼4% of the communication bandwidth.]

K.2 Geodesic Distance Between Communication Subspaces

[1 paragraph: We compare communication matrix subspaces across edges via geodesic distance on the Grassmannian $\mathrm{Gr}(k, d_{\textbf{head}})$. For rank-$k$ subspaces (top-k left singular vectors), we compute principal angles and the geodesic $d_G = \sqrt{\sum \theta_i^2}$. Result: at all ranks ($k = 1, 3, 5, 10$), within-pathway geodesic distances are 85–95% of the maximum possible distance, with across-pathway distances only slightly larger (discrimination ratio 1.03–1.07×). Communication subspaces are near-orthogonal between all edge pairs, regardless of functional relationship.]

[1 paragraph: Interpretation: the geometric structure lives in the *selector subspaces*, not in the communication matrices. Selector subspaces show strong task-specific structure: IOI and SVA route through 88–89° orthogonal subspaces at task-critical layers (Appendix K.4). The communication matrix captures structural *capacity* (the pipe diameter), while the selector subspace captures functional *routing* (which pipe to use). These are complementary views.]

K.3 Factorized vs Standard Composition Scores

[1 paragraph: The factorized and standard paths produce numerically identical scores (geodesic between subspaces < 0.001), confirming that the factorization is faithful.]

[1 paragraph: For cross-edge comparison, individual factor indices show near-zero overlap (Jaccard@10 $= 0.015$ within-pathway) with 8192 factors (∼344 active per channel). However, the discrimination ratio (within/across similarity) is $12.7\times$ for factor Jaccard vs $2.0\times$ for SVD cosine. Full contribution-vector cosine shows within-pathway ∼0.75 vs across-pathway ∼0.66.]

[1 paragraph: Implication: the shared vocabulary operates at the *subspace* level (PCA/Tucker), not at individual factor indices. This motivates the PCA and Tucker decompositions as the natural analysis layer above raw factors.]

K.4 Selector Subspace Orthogonality Across Tasks

[TODO: Redo the Grassmannian orthogonality analysis across all 6 tasks (IOI, SVA, greater-than, hypernymy, capital-country, gender-bias) with proper controls. Preliminary results (v7 slides): top-50 factor Jaccard 0.69–0.85 (shared factors), mean principal angle 75–83° (≈ orthogonal), strengthening to 88–89° at task-critical layers with DST 8192. Tasks share the same factors but route through orthogonal subspaces.]

L Per-Counterfactual Tucker Analysis

L.1 Variance Explained by Counterfactual Type

[1 paragraph: Tucker (10,10,8) variance explained varies dramatically across IOI counterfactual types: s2_io_flip 59.1%, s1_io_flip ∼83.3%, abc ∼52.8%, compound ∼39.5%. s1_io_flip is much more compressible — simpler attribution structure for that corruption type. Compound corruptions have the most complex attribution tensor (least compressible) because they include multiple perturbation types simultaneously.]

L.2 Cross-Counterfactual Mode Comparison

[TABLE — All 6 pairwise comparisons between counterfactual types. Columns: pair, mean matched cosine, mean principal angle, factor Jaccard. Ranked from most similar (s2_io vs compound: 0.440 cos, 58.0°) to most different (s2_io vs s1_io: 0.145 cos, 78.4°). Include interpretation: s2_io and compound share 5/8 modes (cosine >0.64), then 3 nearly orthogonal; s1_io produces structurally simpler tensor with largely distinct modes.]

L.3 Interpretation

[1 paragraph: Different corruption types route through partially distinct factor subspaces. The *shared* modes likely reflect model-intrinsic circuit geometry (always activated regardless of corruption type), while the *distinct* modes reflect corruption-specific information routing. The s1_io_flip case is instructive: swapping the subject (S1) position rather than the indirect object (S2) produces a fundamentally simpler attribution structure (83% vs ∼50% variance explained) with largely distinct modes, suggesting S1 and S2 information flows through different factor subspaces even though both contribute to the same task.]

M Distributed Logit Lens on DAS Dimensions

M.1 Per-Dimension Token Categories

[1 paragraph: Each DAS dimension has a coherent lexical signature when decoded through the unembedding matrix. SVA dim 0 promotes singular verbs (IIA 0.920), dim 1 promotes plural pronouns (IIA 0.960). IOI dims promote names and gender categories. The distributed logit lens reveals what information each causal dimension carries, providing interpretability for the otherwise opaque DAS rotation matrix.]

M.2 Necessity/Sufficiency Taxonomy

[1 paragraph: SVA: 1 privileged direction with 3 redundant copies. IOI: inflated 32-dim subspace; greedy forward search finds 5 optimal dimensions (IIA 0.987) vs full 32 (IIA 0.820, worse due to over-parameterization). Different tasks have qualitatively different causal subspace geometry: SVA is rank-1 with redundancy, IOI is rank-∼5 with padding noise.]

M.3 JS Divergence Clustering

[1 paragraph: JS divergence between token distributions decoded from each DAS dimension reveals semantic clustering. IOI has 3–5 semantically distinct direction types within its 32 DAS dims. Gender Bias has near-duplicate vectors. This clustering suggests the causal subspace has interpretable internal structure that factor-constrained initialization helps discover.]

N Subspace Trimming and Dimensionality Analysis

N.1 IOI Subspace Trimming

[1 paragraph: Greedy forward search: 5 dimensions achieve IIA 0.987 vs full 32-dim DAS at 0.820. $k = 32$ is 16× inflated with -0.167 IIA penalty from noise dimensions that interfere with the intervention. The "optimal" DAS dimensionality is much lower than the trained dimensionality, consistent with the rank-∼5 structure identified by the distributed logit lens.]

N.2 Cross-Task k-Sweep

[1 paragraph: Diminishing returns after $k = 4$ on several tasks. $k = 4$ near-optimal with improved generalization. Larger k overfits to training distribution.]

N.3 Teacher vs Factorized IIA

[1 paragraph: IOI improves on factorized model (0.820 → 0.913 at $k = 32$). SVA/Gender/Capital worsen slightly. Factorization enhances IOI but may subtly damage other tasks — consistent with the factorization creating slightly different internal representations that happen to be better-aligned with IOI's causal structure.]

O Factor Mediation Analysis

[1 paragraph: PC algorithm on factor activations identifies 286 unique factors mediating known IOI composition edges (CI-significant at $\alpha = 0.05$). 90%+ of mediating factors are unique to a single edge; 2 factors capture 50% of mediation. Per-edge factor mediation is highly specific, consistent with the Jaccard overlap results in §5.2.5: the same pattern of factor disjointness observed in attribution scores is replicated by an independent causal discovery algorithm (constraint-based PC) operating on factor activations.]

P Conceptor Steering via Factor Subspaces

[1 paragraph: Boolean operations on DAS-identified factor subspaces: AND (intersection of two task subspaces), OR (union), NOT (complement) produce valid causal interventions at 97–100% IIA. This demonstrates that factor subspace algebra enables compositional steering without retraining DAS. The AND operation identifies shared causal directions between tasks; NOT removes one task's influence while preserving

another's. The factor bank provides the shared basis that makes these operations well-defined.]

Q Comparison to Talking Heads Composition Scores

[1 paragraph: Merullo et al. (2024) compute per-pair SVD of composition scores between head pairs. Our factor bank provides a global basis: the Gram matrix $FF^\top$ mediates all cross-layer communication. Factorized composition via $S_u^\top(FF^\top)S_v$ is algebraically equivalent to standard composition through the shared factor bank. The shared basis enables cross-edge comparison that per-pair SVD cannot — two edges can be compared in the same factor coordinates, whereas per-pair SVD produces incomparable singular vectors.]

R Post-Hoc Sparsity Recovery

[1 paragraph: Seven unsupervised methods tested for recovering selector sparsity post-hoc (without access to the selector matrix): activation thresholding, gradient magnitude, mutual information, correlation analysis, sparse coding, ICA, and spectral methods. All fail — selector sparsity is coordinate-aligned in the learned factor basis and is unrecoverable from activations or gradients alone without supervision. The factorization training procedure is essential; post-hoc analysis cannot substitute. This negative result strengthens the case for distillation-based factorization: the sparse structure must be *trained into* the model, not discovered after the fact.]

S Extended Null Controls

S.1 Random Subspace Controls

[20 random 8D subspaces in $\mathbb{R}^{8192}$: CMD distribution. PCA k=8 CMD should beat all 20 trials by wide margin.]

S.2 Shuffled Factor Controls

[20 shuffled-factor PCA runs: CMD distribution. Histogram with PCA k=8 CMD and full baseline CMD marked.]

S.3 Sub-Variable Orthogonality Null

[1 paragraph: Tucker modes from DAS sub-variable attribution tensors tested against 50 random weightings. Random cosine: 0.08 vs reference; DAS sub-variables: 0.13. Cross-comparison also not significant. Honest null result — the sub-variable Tucker modes do not carry significantly more information than random weightings. This is important to report: it constrains the interpretation of Tucker modes as capturing task-specific (not just data-specific) structure.]

S.4 Factor Bank Projection Null

[1 paragraph: Projecting the factor bank F onto Tucker C subspace produces zero effect at all ranks (k=1,2,4,8). Random subspace projections also produce zero effect. Explanation: F is (8192, 768), so any rank-8 subspace of factor-index space maps to a full-rank-768 output space via the overcomplete basis. Weight-level projection is too weak an intervention; activation-level projection (Section 7.3) is the correct approach.]

T CMD Evaluation of Tucker Factor Subspaces

[TABLE — Tucker CMD comparison across 6 tasks. Per-task Tucker decomposition of each task's own attribution tensor.]

[1 paragraph: Tucker reconstruction beats full EAP-IG only on IOI (0.0178 vs 0.0366). Factor-projected beats random on 5/6 tasks (FP/Rand $<$ 1). greater_than is the sole outlier — Tucker reconstruction catastrophically fails (CMD 0.5164) and random beats factor-projected. This task may have a qualitatively different attribution geometry. High variance capture $\neq$ good edge ranking: SVA achieves 95.7% variance explained but Tucker reconstruction still hurts CMD. Interpretation: Tucker identifies a meaningful factor subspace (not random) but compressing attribution to that subspace generally loses information needed for optimal circuit ranking. IOI's improvement was likely a denoising effect specific to that task's attribution structure.]

U Unsupervised Role Recovery Details

[1 paragraph: PCA on the flattened attribution tensor produces per-edge loading vectors. Clustering these (k-means, $k = 5$ matching the number of Wang et al. functional roles) and comparing to ground-truth role assignments gives NMI = 0.600 (p = 0.004 by permutation test, 1000 permutations). Within-pathway edges have 1.46$\times$ higher cosine similarity of PCA loading vectors than cross-pathway edges (p = 0.004 by bootstrap). "Early NMI" (considering only the first-assigned mode per head) reaches 0.826. The decomposition recovers known functional structure without any supervision, circuit labels, or causal model specification.]

V Figure Gallery

V.1 Per-Edge Factor Fingerprints

Figure 2: **Per-edge factor fingerprints for Ind→SInh edges.** Each row is one edge; columns are factor indices. Binary heatmap showing top-20 factors per edge. Edges between the same head pairs use largely non-overlapping factor subsets — each edge has a unique factor "fingerprint." Generated by `viz_factor_sharing.py`.

Figure 3: **Q/K/V factor sharing for 4 IOI heads.** Per head: Venn-style overlap of top factors used by Q, K, and V projections. Q and K share more factors with each other than with V, consistent with the QK attention-score duality. Generated by `viz_factor_sharing.py`.

Figure 4: **Factor-level discrimination: circuit vs non-circuit factors.** 15.4× discrimination ratio for circuit factors vs 1.5× for non-circuit. Circuit-relevant factors form a statistically distinguishable population. Originally from `artifacts/spectral_results/neurips_figures/`.

V.2 Q/K/V Factor Sharing

V.3 Factor-Level Discrimination

V.4 Factor Overlap Across Circuit Stages

Figure 5: **4×4 Jaccard overlap between IOI circuit stages.** Diagonal: within-pathway Jaccard (0.018–0.037). Off-diagonal: cross-pathway Jaccard (0.001–0.004). Values are mean pairwise Jaccard between different edges within/across the same stage, NOT self-similarity. YlOrRd colormap. Generated by `viz_factor_overlap_v3.py`.

V.5 SAE vs Factor Cross-Layer Tracking

Figure 6: **Cross-layer factor reuse in IOI circuit.** Left panel: 8% of factors participate in 2+ pathway stages at top-20, rising to 17% at top-50. Right panel: example of a factor active across Dup→Ind and Ind→SInh stages. Per-layer SAE features cannot provide this cross-layer tracking. Originally from `artifacts/spectral_results/sae_deep_comparison/exp3_cross_layer_tracking.png`.

V.6 Grassmannian Landscape for DAS Training

Figure 7: **Grassmannian landscape for DAS training on hard IOI,** $\mathrm{Gr}(1, 768)$**.** *Left:* IIA as function of angle to DAS optimal (x) and manifold proximity (y). On-manifold init (blue) follows high-IIA ridge to global optimum (star); off-manifold init (orange) trapped. *Right:* ρ trajectory showing gradient attractor (26%→83%).

Figure 8: **Head-to-head similarity: factor profiles vs edge profiles.** Left: cosine similarity of factor usage profiles (which factors each head uses). Right: cosine similarity of edge profiles (which readers each head connects to). Block structure along diagonal corresponds to Wang et al. functional roles (labeled on left axis). Factor profiles show stronger block structure (higher within-role similarity) than edge profiles. Generated by `viz_25_role_similarity_v2.py`.

Figure 9: **Layer-aggregated PCA sub-circuit heatmaps (IOI, 8 PCs).** Each panel: 13×13 (writer layer $\times$ reader layer). PC0–3: embed/L0 $\rightarrow$ L0 (early processing, 47% variance). PC4–6: L5 $\rightarrow$ L5–L8 (S-inhibition fan-out, 11%). PC7: L9–L11 $\rightarrow$ logits (name movers, 2%). Clean three-tier separation.

Figure 10: **PCA vs Tucker factor direction agreement (IOI).** 8×8 $|\cos\theta|$ matrix = perfect diagonal. All matched pairs have cosine ≥ 0.998, confirming identical factor directions from two mathematically different methods (variance maximization vs tensor reconstruction). Off-diagonal <0.04.

Figure 11: **MDS projection of factor subspace geometry.** Each point is a head; distance reflects Grassmannian distance between factor subspaces. Functional groups (prev-token, dup-token, induction, S-inhibition, name-mover) cluster. Generated by `viz_mds_v4.py`.

Figure 12: **Factor loading heatmap.** Per-head loading on each of the top 8 PCA components. Shows which heads load on which PCs, confirming the functional group assignments. Generated by `viz_factor_loading_heatmap_v2.py`.

V.7 Head-to-Head Role Similarity

V.8 PCA Sub-Circuit Layer Heatmaps

V.9 PCA vs Tucker Factor Direction Agreement

V.10 MDS Subspace Map

V.11 Factor Loading Heatmap

V.12 Cumulative Factor Mass

Figure 13: **Cumulative factor mass for IOI circuit heads.** Each line is one head; x-axis is number of factors (sorted by magnitude); y-axis is cumulative attribution mass. Steep rise confirms sparse factor usage: most heads reach 80% mass with <50 factors (of 8192). Generated by `viz_head_factor_fingerprints.py`.

Figure 14: **Pairwise Jaccard similarity between IOI heads (top-20 factors).** Block-diagonal structure matches Wang et al. roles. Within-role Jaccard $\sim$0.1–0.2; across-role Jaccard $<$0.05. Different from the stage-level Jaccard (Figure V.4) which pools edges within pathways. Generated by `viz_head_factor_fingerprints.py`.

Figure 15: **Factor subspace evolution across layers.** Grassmannian distance between layer-l and layer-$l+1$ factor subspaces for IOI circuit heads. Large jumps at L5 and L9 correspond to functional transitions (early processing $\rightarrow$ S-inhibition $\rightarrow$ name movers).

Figure 16: **Hierarchical clustering of all 144 IOI circuit edges by factor attribution profile.** Dendrogram with functional role coloring. Edges from the same functional group cluster together (NMI = 0.600). The method discovers the circuit's role structure without supervision. Generated by `viz_batch3.py`.

V.13 Head Pairwise Jaccard Heatmap

V.14 Cross-Layer Factor Subspace Evolution

V.15 Full Edge Clustering

V.16 CMD Method Comparison

Figure 17: **CMD comparison across decomposition methods.** PCA, Tucker, Varimax, ICA, PLS-DA at various ranks. PCA and Varimax dominate; ICA substantially worse. Generated by `viz_cmd_method_comparison.py`.

V.17 Scree Plot and Cumulative Variance

Figure 18: **Scree plot for IOI attribution tensor PCA.** Primary/secondary gap at $k = 4$; secondary/tail transition at $k = 8$. Auto-k at 90% variance selects $k = 108$.

V.18 Cumulative Variance by Task

Figure 19: **Cumulative variance explained by PCA rank, per task.** SVA reaches 90% at $k = 5$ (simple circuit); Gender requires $k = 190$ (complex). The auto-k threshold adapts to circuit complexity.

V.19 DAS Direction Angles

V.20 IIA vs CMD Scatterplot

Figure 20: **Pairwise cosine between PCA components and DAS directions.** Off-diagonal, confirming the $\sim 88°$ orthogonality between attribution and causal subspaces.

Figure 21: **IIA vs CMD across methods and tasks.** Shows the tradeoff between causal accuracy (IIA) and circuit faithfulness (CMD). Points colored by method. PCA-projected factorized EAP achieves the best CMD; trained DAS achieves the best IIA.